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Soil and Rock Logging Explanatory Notes and Abbreviations

AS1726-2017

Soil and rock descriptions on the engineering log sheets are in accordance with AS1726 – 2017-Geotechnical Site Investigations. For full details of the logging standards refer to Section 6 of AS1726-2017. A summary of the sequence of descriptive terms used to describe soil and rock on the engineering log sheets is outlined below.

AECOM logs prepared prior to July 2017 were prepared generally in accordance with AS1726 - 1993. The soil classification system adopted in AS1726-2017 differs from the Unified Soil Classification System (USCS) and the logging standards recommended in AS1726-1993 significantly in the criteria used to distinguish between coarse and fine grained soils (see Soil Description below).

Soil Description

SOIL NAME: plasticity or particle characteristics of major component; colour; structure; secondary and other minor components. The AS1726 Group Symbol; consistency/density; and moisture condition are listed as abbreviations in separate columns. Geological origin and additional observations as required such as soil origin i.e. FILL, ALLUVIUM and other significant details are recorded in a separate column.

The major soil component (capitalised) is assessed after removal of cobble size or larger components (>63mm), soils containing 35% or more passing the 75 micron sieve are classified as fine grained soils. Fine grained soils are classified on the behaviour of the fine grained component, not percentage of silt and clay.

This is a significant departure from the previous AS1726-1993 where soils had to contain 50% or more passing the 75 micron sieve to be classified as a fine grained soil. This results in materials containing 50% or more coarse grained material previously described as coarse grained (e.g. "sandy CLAY/clayey SAND" or "silty SAND") now being described as fine grained soils if they contain at least 35% fines (e.g. "sandy CLAY" or "sandy SILT"). Additionally, many fine grained soils previously described as containing both silt and clay (e.g. "silty CLAY", "CLAY: with silt") are now described as either CLAY or SILT with no mention of the other fine component. The exception is where behaviour is borderline, in which case "clayey SILT" is used.

Rock Description

ROCK NAME: grain size and type; colour; fabric and texture; structure; minor components; and bedding dip. The geological formation; rock strength; weathering/alteration; mass defect spacing; and defect descriptions are listed under separate columns.

Extremely weathered material is prefixed as such followed by a description of the soil properties. Similarly in situ weathering or infilling of defects greater than 100 mm thick with soil or extremely weathered material is described using relevant soil properties in the description column of the engineering log sheet.

Field Samples and In Situ Tests

Field samples and tests are recorded in the relevant column using abbreviations described under General Symbols and Abbreviations.

Field tests have been used to assess soil consistency/density and rock strength, and unless specifically stated otherwise, have been transferred directly to the engineering log sheets and not modified to coincide with laboratory results. Field descriptions may be used as an independent estimate of material properties which can be correlated with other data.

Organic and Artificial Material

Soil containing greater than 25% organic material is described as PEAT. Soils containing between 2% and 25% organic material are prefixed by "Organic" (e.g. Organic CLAY). TOPSOIL is a specific soil origin that may also contain both living and dead organic materials. Organic matter is described using terms such as fibrous peat, charcoal, wood fragments, roots (>2mm diameter) or root fibres (<2mm diameter)

Any material containing evidence that it has been placed by man-made mechanisms, including compacted embankment materials and materials containing evidence of artificial materials is prefixed as FILL. Waste fill is described using terms such as domestic refuse, oil, bitumen, brickbats, concrete rubble, fibrous plaster, wood pieces, wood shavings, sawdust, iron filings, drums, steel bars, steel scrap, bottles, broken glass, or leather.

Organic and artificial material cannot be adequately described using soil classification terms. They are mentioned, at the end of the description using qualitative terms such as "rare", "occasional" or "frequent", e.g. "SAND with rare gravel size brick fragments". These qualitative terms are relative, for which no definition of percentage is given.

Structure

The structure of soil (or rock) is usually applicable to cohesive soils or rock. Typical terms that are used on the engineering log sheets include*; intact* (no joints), *fissured* (closed joints), *voided*, *vesicular*, *slickensided* (sheared), *interbedded***,** *laminated* and *cemented*.

Plasticity

Descriptive Term	Range of Liquid Limit (%)
Low plasticity	≤ 35
Medium plasticity	> 35 ≤ 50
High plasticity	> 50

Colour

Colour has been assessed in the "moist" condition using basic colours and the modifiers pale, dark and mottled. Borderline colours are described as a combination of the two colours (e.g. red-brown). When describing the colour of defect infill, the following abbreviations are used in the defect description column.

Colour abbreviations

Term	Abbreviation
Brown	br
Grey	gy
Black	bk
White	wh
Blue	bl
Green	gr
Yellow	yl
Orange	or
Red	rd
Pale	pl
Dark	dk
Mottled	mtld

Moisture Condition

Term	Symbol	Description	
		Cohesive	Granular
Dry	D	Cohesive; hard and friable or powdery, very dry of plastic limit	Cohesion-less and free running
Moist	M w<PL w~PL w>PL	Soil feels cool, darkened in colour, can be moulded, typically with moisture content (w) between plastic limit (PL) and liquid limit (LL) as indicated	Soil feels cool, darkened in colour, tends to cohere
Wet	W w~LL w>LL	Soil feels cool, dark, usually weakened, free water, typically with moisture content (w) approximately at or wet of liquid limit (LL) as indicated	Soil feels cool, darkened in colour, tends to cohere, free water

Geological Origin

Term		Description
Weathered in Place Material	Extremely weathered material	Structure and fabric of parent rock visible (NOTE – described with soil terminology)
	Residual Soil	Structure and fabric of parent rock not visible
Transported Soils	Aeolian soil	Deposited by wind
	Alluvial soil	Deposited by streams and rivers
	Colluvial soil	Deposited on slopes (transported downslope)
	Lacustrine soil	Deposited by lakes
	Marine soil	Deposited in oceans, bays, beaches and estuaries
TOPSOIL	Mantle of surface and/or near surface soil often but not always defined by high levels of organic material (prefix to soil description)	
FILL	Soil, rock or refuse placed by humans in either controlled or uncontrolled conditions (prefix to soil description)	

Where there is doubt to origin 'possibly' or 'probably' are used.

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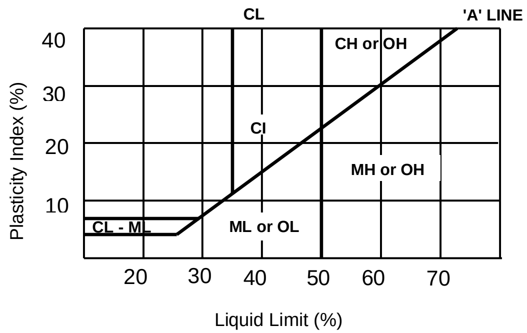
Soil Classification

Field Identification Procedures (Excluding particles larger than 63mm and basing fractions on estimated mass)				Log Graphic	Group Symbol	Typical Names	Secondary and Minor Components		Laboratory Classification Criteria
Coarse Grained Soils More than 65% of material less than 63 mm is larger than 0.075 mm	GRAVEL More than 50% of coarse fraction is larger than 2.36mm	clean GRAVEL (<5% fines)	Wide range in grain size and substantial amounts of all intermediate sizes, not enough fines to bind coarse grains, no dry strength		GW	Well graded GRAVEL or sandy GRAVEL	Secondary/Minor Descriptor Prefix "sandy", "silty" etc. Add "with sand/silt" etc. Add "trace sand/silt" etc. Coarse Grains >30% >15%,≤30% ≤15% Fine Grains >12% >5%,≤12% ≤5%	Laboratory grain size curve used in identifying the fractions as given under field identification	$c_u = \frac{D_{60}}{D_{10}}$ Cu ≥4 $c_c = \frac{(D_{30})^2}{D_{10} x D_{60}}$ Cc=1 – 3
			Predominantly one size or range of sizes with some intermediate sizes missing, not enough fines to bind coarse grains, no dry strength		GP	Poorly graded GRAVEL or sandy GRAVEL			<5% fines, not meeting all gradation requirements for GW.(i.e. Cu < 4 or Cc ≠ 1 – 3)
		GRAVEL (≥12% fines)	'Dirty' materials with excess of non-plastic fines, zero to medium dry strength		GM	silty GRAVEL or silty sandy GRAVEL			≥12% Fines with Atterberg limits below 'A' line
			'Dirty' materials with excess of plastic fines, medium to high dry strength		GC	clayey GRAVEL or clayey sandy GRAVEL			≥12% Fines with Atterberg limits above 'A' line
	SAND More than 50% of coarse fraction is smaller than 2.36mm	Clean SAND (<5% fines)	Wide range in grain size and substantial amounts of all intermediate sizes, not enough fines to bind coarse grains, no dry strength		SW	Well graded SAND or gravelly SAND			$c_u = \frac{D_{60}}{D_{10}}$ Cu ≥6 $c_c = \frac{(D_{30})^2}{D_{10} x D_{60}}$ Cc=1–3
			Predominantly one size or range of sizes with some intermediate sizes missing, not enough fines to bind coarse grains, no dry strength		SP	Poorly graded SAND or gravelly SAND			<5% fines, not meeting all gradation requirements for SW. (i.e. Cu < 6 or Cc ≠ 1 – 3)
		SAND (≥12% fines)	'Dirty' materials with excess of non-plastic fines, zero to medium dry strength		SM	silty SAND			≥12% Fines with Atterberg limits below 'A' line.
			'Dirty' materials with excess of plastic fines, medium to high dry strength		SC	clayey SAND			≥12% Fines with Atterberg limits above 'A' line.

Fine Grained Soils 35% or more of material less than 63 mm is smaller than 0.075 mm	IDENTIFICATION PROCEDURES ON FRACTIONS < 0.2 mm				Log Graphic	Group Symbol	Typical Names
	LIQUID LIMIT	DRY STRENGTH	DILATANCY	TOUGHNESS			
SILT	>50%	Low to medium	None to slow	Low to medium		MH	Inorganic SILT, gravelly SILT, sandy SILT
	≤50%	None to low	Slow to rapid	Low		ML	Inorganic SILT and very fine sand, rock flour, silty or clayey fine sand or silt.
		Low to medium	Slow	Low		OL	Organic SILT (2%<organics<25%)
Borderline SILT/ CLAY	≤35%	Low to Medium	None to slow	Low to Medium		ML-CL	clayey SILT
CLAY	>50%	High to very high	None	High		CH	Inorganic CLAY, sandy CLAY, gravelly CLAY
	>35 and ≤50%	Medium to high	None to slow	Medium		CI	
	≤35%					CL	
	>35%	Medium to high	None to very slow	Low to medium		OH	Organic CLAY (2%<organics<25%)
PEAT (>25% organic by dry weight)		Readily identified by colour, odour, spongy feel and frequently by fibrous texture				Pt	PEAT, sandy PEAT

Secondary and Minor Components: Secondary fine grained soil not described except for borderline SILT/CLAY mixtures described as "clayey SILT". Coarse grained Secondary/Minor descriptors: >30% prefix "gravelly" or "sandy" as appropriate; >15% ≤30% add "with sand/gravel" as appropriate; ≤15% add "trace sand/gravel" as appropriate.

Laboratory Classification Criteria: Laboratory classification from Modified Cassagrande Chart



Modified Cassagrande Chart for classification of fine-grained soils

Boundary classifications – Soils possessing characteristics of two groups are designated by combinations of group symbols. For example: GP-GC for gravel with between 5 and 12% clay fines; CI-CH for clay soils where field estimates of plasticity are borderline between medium and high

Other log graphics: FILL, TOPSOIL, ASPHALT, CONCRETE, sandy GRAVEL, gravelly SAND, gravelly SILT, sandy SILT, gravelly CLAY, sandy CLAY

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Sedimentary and Metamorphic Rock Types, Grain Size, Defect Spacing and Planar Structure

Grain Size/ Spacing Thickness	Soil Grain Size Term	Rock Type							Defect Spacing Term	Bedding Thickness Term
		Sedimentary				Metamorphic		Igneous/ Metamorphic grain size		
		Deposited rock type	≥90% carbonate (prefix IMPURE 50%-90%)		Volcanic ejecta	Foliated	Non-Foliated			
			Low porosity	Porous						
> 2m	large BOULDERS	CONGLOMERATE (larger rounded grains in a finer matrix) *or* BRECCIA (angular or irregular rock fragments in a finer matrix)	LIMESTONE (predominantly $CaCO^3$) or DOLOMITE (predominantly $CaMgCO^3$)	CALCIRUDITE	AGGLOMERATE (rounded grains in a finer matrix) *or* VOLCANIC BRECCIA (angular or irregular fragments in a finer matrix)	GNEISS	MARBLE (crystalline $CaCO^3$)/ QUARTZITE (fused quartz)/ SERPENTINITE (mafic igneous)/ HORNFELS (fine thermal)	COARSE	VERY WIDE (VW)	VERY THICKLY BEDDED
0.6 - 2m	medium BOULDERS								WIDE (W)	THICKLY BEDDED
0.2 - 0.6 m	small BOULDERS								MEDIUM (M)	MEDIUM BEDDED
60 - 200 mm	COBBLES								CLOSE (C)	THINLY BEDDED
20 - 60 mm	coarse GRAVEL								VERY CLOSE (VC)	LAMINATED
6 - 20 mm	medium GRAVEL								EXTREMELY CLOSE (EC)	THINLY LAMINATED
2 - 6 mm	fine GRAVEL								-	VERY THINLY LAMINATED
0.6 - 2.0 mm	coarse SAND	SANDSTONE/ GREYWACKE (rock fragments)/ ARKOSE (mainly feldspar)/ QUARTZOZE SANDSTONE (quartz and siliceous cement)		CALCARENITE	TUFF	SCHIST		MEDIUM		-
0.2 - 0.6 mm	medium SAND									
0.06 - 0.2 mm	Fine SAND									
2 - 60 µm	SILT	MUDSTONE/ SHALE (fissile)/ LAMINITE (bedded); SILTSTONE		CALCISILTITE	Fine grained TUFF	PHYLLITE (undulose)/ SLATE (planar)		FINE		
< 2 µm	CLAY	CLAYSTONE		CALCILUTITE						

Note: EVAPORITE is sedimentary rock which consist mainly of salts e.g.: Halite (NaCl), Anhydrite and Gypsum ($CaSO_4$); COAL is mostly organic sedimentary rock that consists of indurated accumulations of plant debris

Classification of Igneous Rocks (Massive Crystalline)

Grain Size (mm)	Much quartz, pale (felsic)	⟷	Little quartz, dark (mafic)
COARSE (>2)	GRANITE	DIORITE	GABBRO
MEDIUM (0.06-2)	MICROGRANITE	MICRODIORITE	DOLERITE
FINE (<0.06)	RHYOLITE	ANDESITE	BASALT

Note:
- PEGMATITE consist of large crystals often forming a dyke or vein
- VOLCANIC GLASS and OBSIDIAN are rocks that have cooled quickly and have a glassy texture
- APLITE may occur as light coloured veins of quartz and feldspar in other igneous rocks
- PORPHYRY is an igneous rock consisting of large crystals in a much finer matrix

Duricrust (soils cemented to rock)

FERRICRETE (iron oxide cemented)
SILCRETE (silica cemented)
GYPCRETE (salt cemented)
CALCRETE ($CaCO_3$ cemented – dominated by replacement features)

No Core

NO CORE (description, defect and strength logs show beak in logging)

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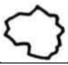
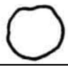
Grain Size

Soil Type	Fine Grained Soil		Coarse Grained Soil						Oversize
	Clay	Silt	Sand			Gravel			Cobbles
			Fine	Medium	Coarse	Fine	Medium	Coarse	
Grain Size	< 2 µm	2-75 µm	0.075-0.21 mm	0.21-0.6 mm	0.6-2.36 mm	2.36-6.7 mm	6.7-19 mm	19-63 mm	63-200 mm
Field Guide	Shiny, Not visible under 10x	Dull, Visible under 10x	Visible by eye	Visible at < 1 m	Visible at < 3 m	Visible at < 5 m	Road gravel	Rail ballast	Beaching

Boulders (>200mm) and Cobbles (63-200mm) are considered oversize materials and are separated from the sample prior to describing the soil. Where cobbles or boulders are present the soil description is preceded by 'MIXTURE OR SOIL AND COBBLES/BOULDERS' with the word order indicating the dominant proportion first followed by the soil description and the estimate proportion of oversize materials indicating whether they are supported by the soil matrix.

Grain Shape

	Angular	Sub-angular	Sub-rounded	Rounded
Equi-dimensional particles				

Essentially two-dimensional particles are described as "flaky" or "platy" while essentially one dimensional particles are described as "elongated". Particle composition should be described where significant (e.g. "quartz sand")

Relative Density (non-cohesive soils)

Term	Very Loose	Loose	Medium Dense	Dense	Very Dense
Symbol	VL	L	MD	D	VD
Density Index (%)	≤ 15	> 15 ≤ 35	> 35 ≤ 65	> 65 ≤ 85	> 85
Uncorrected SPT (N) Blow count #	0 - 4	4 - 10	10 - 30	30 - 50	> 50
Field Guide	Ravels	Shovels easily	Shovelling very difficult	Pick required	Pick difficult

NOTE: Density Index as a measure of relative density only applies to dry cohesionless soils. Correlations between relative density and SPT vary considerably depending on grain size and angularity, overburden pressure, moisture content, fines content, cementation and SPT efficiency. The above SPT correlation is a rough field guide for clean, dry, fine to medium sands at a depth of 5 to 15m after Gibbs & Holtz (1957). Where relative density is important to design, more detailed correlations taking into account the factors discussed above should be considered.

Consistency (cohesive soils)

Based on undrained strength (S_u) estimated in field from pocket penetrometer or shear vane

Term	Very Soft	Soft	Firm	Stiff	Very Stiff	Hard
Symbol	VS	S	F	St	VSt	H
Undrained Shear Strength (kPa)	≤12	>12 ≤ 25	>25 ≤ 50	>50 ≤100	>100 ≤ 200	>200
Approx. SPT (N) Blow count #	0 - 2	2 - 4	4 – 8	8 - 15	15 - 30	> 30
Field Guide	Exudes between the fingers when squeezed	Can be moulded by light finger pressure	Can be moulded by strong finger pressure	Cannot be moulded by fingers. Can be indented by thumb	Can be indented by thumb nail	Can be indented with difficulty with thumb nail

NOTE: SPT is not a direct measure of consistency and ranges are only a rough field guide. Plasticity, fissuring, moisture, and sand and gravel content can all affect SPT values dramatically.

Rock Strength

Term	Soil	Very Low	Low	Medium	High	Very High	Extremely High
SYMBOL	soil	VL	L	M	H	VH	EH
UCS (MPa)	≤ 0.6	> 0.6 ≤ 2	>2 ≤ 6	> 6 ≤ 20	> 20 ≤ 60	> 60 ≤ 200	> 200
$I_{s\,(50)}$ (MPa) see note 1&3	-	> 0.03 ≤ 0.1	> 0.1 ≤ 0.3	> 0.3 ≤ 1	> 1 ≤ 3	> 3 ≤ 10	> 10
FIELD GUIDE	Soil strength – logged as soil using consistency	Material crumbles under firm blow with sharp end of pick. Can be peeled with a knife. Too hard to cut a triaxial sample by hand. Pieces up to 3 cm thick can be broken by finger pressure.	Easily scored with a knife. Indentations of 1mm - 3mm in the specimen with firm blows of the pick point. Has dull sound under hammer. A piece of core 150 mm long 50 mm diameter may be broken by hand	Readily scored with a knife. A piece of core 150 mm long 50 mm diameter can be broken by hand with difficulty	A piece of core 150 mm long 50 mm diameter cannot be broken by hand but can be broken with by a pick with a single firm blow. Rock rings under hammer	Hand specimen breaks with a pick after more than one blow. Rock rings under hammer	Specimen requires many blows with a geological pick to break through intact material. Rock rings under hammer

Note:
1. The strength of rock material should be based on UCS of materials close to in situ moisture content. $I_{s(50)}$ should only be used where not practical to undertake UCS tests
2. Anisotropy of rock material samples may affect the field assessment of strength
3. The unconfined compressive typically ranges from 10 to 20 times the $I_{s(50)}$ but the multiplier may vary widely for different rock types

Degree of Weathering / Alteration

Weathering and alteration relate to the rock fabric. Both involve physical and chemical changes to the rock due to changes in pressure, temperature, moisture and chemical environment. Weathering is in response to changes from exposure at the earth's surface while alteration is caused by hot liquids or gasses at depth. The distinction between weathering and alteration is important because they are likely to have different distribution patterns.

Degree of Weathering / Alteration		Weathering Symbol/ Alteration Symbol		Weathering/ Alteration Description
Residual Soil		RS		Material is weathered to such an extent that it has soil properties. Mass structure and material texture and fabric of original rock are no longer visible but the soil has not been significantly transported.
Extremely Weathered / Extremely Altered		XW / XA		Material is weathered/ altered to such an extent that it has soil properties. Mass structure and material texture of original rock are still visible. Logged as "Extremely weathered/ altered material" and described as a soil.
Highly Weathered / Highly Altered	Distinctly Weathered/ Distinctly Altered	HW / HA	DW/ DA	The whole of the rock material is discoloured usually by staining or bleaching to the extent that the colour of the original rock is not recognisable. Rock strength is changed by weathering/alteration. Some primary minerals weathered to clay minerals. Porosity may be increased by leaching, or may be decreased due to deposition of weathering product/precipitation of secondary minerals in pores.
Moderately Weathered / Moderately Altered		MW / MA		The whole of the rock material is discoloured usually by staining or bleaching to the extent that the colour of the original rock is not recognisable but shows little or no change in strength from fresh rock.
Slightly Weathered / Slightly Altered		SW / SA		Rock is partially discoloured (with staining or bleaching along joints in the case of weathering) but shows little or no change in strength from fresh rock.
Fresh		FR		Rock shows no sign of decomposition of individual minerals or colour changes.

NOTE:
1. The term "Extremely Weathered/Altered rock" as a description is misleading. Described as Extremely weathered/altered material" followed by material description as a soil.
2. DW/ DA = Distinctly Weathered/Altered is used where it is not practical to distinguish between HW/HA and MW/MA on basis of strength change

In some instances it may be appropriate to describe weathering as part of a rock mass classification system. Where this is used the rock mass classification including the specifics of the weathering related to each rock class is described in the body of the report.

Common Defects in Rock Masses

Defects are described in the description column in the following order, defined by abbreviations:

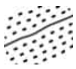
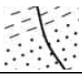
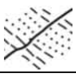
Type; dip/direction; planarity; roughness; infill/coating; colour. To indicate the defect has been healed, healed is printed at before the defect type.

E.g. P,30/145°,PL,ro,1mm,CH,gy indicates a parting with 30° dip, 145° dip direction, planar, rough surfaces, 1mm thick, filled with grey high plasticity clay.

Defects with a perpendicular thickness up to 10 mm thick are described as partings or joints. Defects 10mm to 100mm thick perpendicular to the defect are described as seams or zones. Defects greater than 100mm thick or intersecting the full width of the core for more than 100mm are described as new material strata on the log.

Defects can be either described individually or grouped into generalised defect sets with similar properties. Individual defects are marked with a leader on the left side of the defect portion of the log, while generalised defects are written on the right side of the defect description area of the log. The extent to which defects are described individually or generalised depends on the nature of the core being logged.

Defect Type

Log Symbol		Appearance		Term	Definition
P				**Parting**	A surface or crack, parallel or sub-parallel to layering or planar anisotropy (bedding or cleavage), across which the rock has little or no tensile strength. May be open or closed
J				**Joint**	A surface or crack, with no apparent shear displacement across which the rock usually has little tensile strength, but which is not parallel or sub-parallel to bedding/cleavage.
S				**Sheared Surface#**	A near planar, curved or undulating surface which is usually smooth, polished or slickensided and which shows evidence of shear displacement.
SZ				**Sheared Zone#**	Zone of rock material with roughly parallel near planar, curved or undulating boundaries cut by closely spaced joints, sheared surfaces or other defects. Some of the defects are usually curved and intersect to divide the mass into lenticular or wedge shaped blocks.
MB				**Mechanical Break**	A break in rock mass not caused by natural effects. Example causes include drilling, testing and storage
SH			**Seams**	**Sheared Seam#**	Seam of roughly parallel boundaries of rock substance cut by closely spaced joints or cleavage surfaces.
CR				**Crushed Seam#**	Seam with roughly parallel boundaries composed of mainly angular fragment of the host rock substance.
NF	SS*			**Infilled Seam**	Seam with distinct roughly parallel boundaries. The infill is caused by migration of soil into open joints.
EW				**Extremely Weathered Seam**	Seam of soil substance weathered from host rock.

NOTES: # sheared, surfaces, sheared zones, sheared seams and crushed seams are generally faults in geological terms.
* SS = Soil Seam where not possible to determine origin
Healed defects are preceded by "healed"

Defect Planarity

Symbol	Description
PL	planar
CU	curved
UN	undulating
ST	stepped
IR	irregular

NOTE: where large scale waviness is observed it should be further described

Defect Roughness

Symbol	Description
vr	Very Rough
ro	Rough
sm	Smooth
po	Polished
sl	Slickensided

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Infill/Coating

Symbol	Description
cn	clean
sn	stained
vn	veneered
co	coated
op	open/voided
Ca	Calcium Carbonate
Fe	Iron Oxide
Ch	Chlorite
Qz	Quartz

Vesicularity

Symbol	Description	Porosity
D	Dense	Negligible
NV	Non-vesicular	< 10%
SV	Slightly vesicular	10 - 20%
HV	Highly vesicular	> 20%

Cementation and Duricrust

'Weakly cemented' soils are easily disaggregated by hand in air or water.

'Moderately cemented' soils require effort to disaggregate the soil by hand in air or water.

Consistent cementation throughout the material is classified as a 'Duricrust' and logged as a rock

Duricrust Mass Grade

Grade	Term	Description
DI	Massive or Hardspan	>90% duricrust rock forms continuous framework
DII	Vuggy or Patchy	50-90% duricrust rock forming continuous framework around soil(vuggy) or rock (patchy) materials
DIII	Nodular or Fragmental	<50% cemented gravel or cobble size nodules or fragments within soil (described as soil)

Carbonate Soils and Rocks

Rocks and soils with less than approximately 50% carbonate content as indicated by weak or sporadic effervescence in the presence of 10% HCl solution are prefixed 'Calcareous'.

Soils with >50% carbonate are prefixed 'Carbonate'.

Sedimentary Rocks comprising 90% or more carbonate are described using the appropriate rock type name (see rock description). Where carbonate content is 50% to <90%, the rock type is prefixed by 'IMPURE'.

Field Sampling and Testing Abbreviations

Symbol	Description
V	Uncorrected Borehole Vane Shear (kPa) – Peak/Residual
HV	Uncorrected Hand Vane Shear (kPa) – Peak/Residual
PP	Pocket Penetrometer (kPa)
SPT	Standard Penetration Test
N	Uncorrected SPT blow count for 300 mm
N*	SPT with sample collected
RW	SPT rod weight only (SPT N < 1)
HW	SPT rod and hammer weight (SPT N < 1)
HB	SPT Hammer Bouncing
FPM	Field Permeability
Lu	Lugeon/Packer Test (L/m/min)
$Is_{(50)}$(A)	Axial Point Load Strength Index (MPa)
$Is_{(50)}$(D)	Diametral Point Load Strength Index (MPa)
$Is_{(50)}$(I)	Irregular Point Load Strength Index (MPa)
U(X)	Undisturbed Sample (X) mm diameter
UP	Undisturbed Piston Sample
DS	Disturbed Sample
BS	Bulk Sample
E	Environmental Sample
RQD	Rock Quality Designation (%)

Symbol	Description
SCR	Solid Core Recovery (%)
TCR	Total Core Recovery (%)
DCP	Dynamic Cone Penetration Resistance (blows/100 mm)
PSP	Perth Sand Penetrometer Resistance (blows/ 150 mm)
PID	Photoionization Detector

Water

Symbol	Description
▼	Water level (static)
▽	Water level (during drilling)
▷	Water inflow
◁	Water outflow
◀	Complete water loss

Drilling Method

Drilling Method Symbol	Description
ADV	Auger Drilling – V-Bit (100mm)
AS	Auger Screwing
WB	Wash Boring
B	Blank Bit*
T	Tungsten Carbide Bit*
RR	Rock Roller/Tricone
DHH	Down Hole Hammer
PD	Percussion Drilling
CT	Cable Tool
HA	Hand Auger
DT	Diatube (114mm)
NMLC	NMLC Size Core – Triple Tube (50mm diameter)
NQ3, HQ3, PQ3	Wireline Size Core – Triple Tube (45mm, 61mm, 83mm diameter)
NQ, HQ, PQ	Wireline Size Core – Double Tube (48mm, 64mm, 85mm diameter)
RC	Reverse Circulation
CA	Casing Advancer
VC	Vibro Coring
SC	Sonic Coring
GP	Geoprobe Continuous Sampling

*Drill bit symbol used as suffix to drilling method symbol, e.g. ADV indicates auger drilling with V-bit

Drilling Support

Symbol	Description
U	Unsupported
C	Casing
M	Mud
W	Water

DRAFT

Engineering Log

Borehole No. ABH1/MW008

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 29/04/26
End Date: 30/04/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 13.05 m
Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A
Easting: 338457.54
Northing: 6254468.30
Hor. Datum: MGA2020-56
RL: 26.1 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	B: 0.00-0.30 m		26.0	0.0			TOPSOIL - Clayey SAND: fine to medium grained, dark brown; trace gravel, fine to medium, sub-rounded, sandstone .	M		TOPSOIL
								FILL - Clayey SAND: fine to medium grained, brown mottled yellow; trace gravel, fine to coarse, sub-rounded, sandstone ; trace cobbles, size up to 64mm, sub-rounded, sandstone .	M		FILL
				25.0	1.0						
				24.0	2.0						
				23.0	3.0						
				22.0	4.0						
					5.0						

Log continued on next page.

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DRAFT

AECOM Engineering Log

Cored Borehole No. ABH1/MW008

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 29/04/26
End Date: 30/04/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 13.05 m
Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A
Easting: 338457.54
Northing: 6254468.30
Hor. Datum: MGA2020-56
RL: 26.1 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
				26.0	0.0		*Log continued from previous page.*					
NMLC	1						FILL - MASONRY: pale grey, with visible voids; bricks with mortar joints.		26 (15) [0]			
				25.0	1.0		NO CORE: from 0.50 to 1.10m.					
	1.10 / 2		C: 1.28-1.39 m Is(50) D=3 MPa; C: 1.28-1.39 m Is(50) A=1.9 MPa				FILL - MASONRY: brown to reddish brown; bricks with mortrr joints.		38 (32) [0]			
				24.0	2.0		NO CORE: from 1.52 to 2.20m.					
	2.20 / 3		C: 2.20-2.31 m Is(50) D=0.46 MPa; C: 2.20-2.31 m Is(50) A=0.99 MPa; C: 3.13-3.23 m Is(50) D=1.8 MPa; C: 3.13-3.23 m Is(50) A=0.34 MPa	23.0	3.0		FILL - MASONRY: grey to reddish brown; bricks with mortar joints.		78 (78) [0]			
				22.0	4.0		NO CORE: from 3.77 to 4.20m.					
	4.20 / 4		C: 4.63-4.73 m Is(50) D=0.6 MPa; C: 4.63-4.73 m Is(50) A=0.53 MPa				FILL - MASONRY: reddish brown; bricks with mortar joints.		48 (42) [0]			
					5.0		NO CORE: from 4.83 to 5.50m.					

DRAFT

AECOM Engineering Log

Cored Borehole No. ABH1/MW008

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 29/04/26
End Date: 30/04/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 13.05 m
Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A
Easting: 338457.54
Northing: 6254468.30
Hor. Datum: MGA2020-56
RL: 26.1 m
Ver. Datum: AHD
Surface: Topsoil

Method: NMLC. Ground water: 08/05/2026 13:07:43.

Core Run	Samples & Field Tests	RL (m AHD)	Depth (m)	Material Description	Weathering	TCR (SCR) [RQD] (%)	Additional Observations (Defect Descriptions)
		21.0		NO CORE: from 4.83 to 5.50m.			
5.50				ROOT: black.		100 (69) [38]	
5	C: 5.79-5.87 m Is(50) D=0.16 MPa; C: 5.79-5.87 m Is(50) A=0.28 MPa	20.0	6.0	SANDSTONE: fine to coarse grained, brown, indistinctly bedded.	MW		HAWKESBURY SANDSTONE 5.77 m: P, 45°, RF, PR, CT wood 5.88 m: P, 5°, RF, PR, CN 5.90 m: J, 55°, RF, PR, CT clay 5.95 m: P, 0°, RF, PR, SN clay 6.00 m: MB 6.11 m: P, 5°, RF, PR, SN clay 6.21 m: P, 5°, RF, PR, CT clay
				SANDSTONE: medium to coarse grained, red-brown mottled grey, indistinctly bedded.	HW / XW		6.31-6.36 m: P, 5°, RF, PR, CN, 30mm spacing, X3, healed 6.38 m: P, 45°, RF, PR, CT clay 6.44 m: J, 85°, RF, PR, VN clay, 0.1 m persistence 6.45 m: P, 30°, RF, PR, CN 6.51 m: P, 45°, RF, PR, CT clay 6.52-6.70 m: CS, 30°, RF, PR, VN clay, 180mm
	C: 6.75-6.84 m Is(50) D=0.66 MPa; C: 6.75-6.84 m Is(50) A=0.83 MPa	19.0	7.0	SANDSTONE: fine to medium grained, pale grey, cross-bedded at 0-15°, with dark grey laminations .	SW		6.76 m: P, 10°, RF, PR, SN Fe 7.00 m: MB 7.03 m: MB
7.20						100 (96) [96]	7.20 m: DB 7.63 m: HB
6	C: 7.79-7.88 m Is(50) D=0.77 MPa; C: 7.79-7.88 m Is(50) A=1.1 MPa	18.0	8.0	SANDSTONE: fine with medium grained, brown-grey, with white quartz inclusions, sub-rounded, size up to 5mm. From 8.26m, orange brown, iron stained	MW		7.87 m: MB 8.00 m: MB
	C: 8.18-8.27 m Is(50) D=0.14 MPa; C: 8.18-8.27 m Is(50) A=0.1 MPa						8.18 m: MB
				SANDSTONE: fine to medium grained, pale grey mottled brown, cross-bedded at 0-20°, with dark grey laminations .	XW / SW		8.33-8.40 m: EW, 5°, RF, PR, VN clay, 70mm thick
8.74						100 (100) [96]	8.74 m: DB 8.90 m: DB
	C: 9.00-9.20 m UCS=22 MPa; C: 9.20-9.29 m Is(50) D=0.94 MPa; C: 9.20-9.29 m Is(50) A=1.1 MPa	17.0	9.0				9.00 m: DB
			10.0				9.94 m: DB

Inferred Strength: 20 x Is(50) A: ● D: o UCS: □ (SOIL, VL, L, M, H, VH, EH; 0.6, 2, 6, 20, 60, 200)
Defect Spacing (mm): 20, 60, 200, 600, 2000

DRAFT

AECOM Engineering Log

Cored Borehole No. ABH1/MW008

Client: Department of Defence		**Project No:** 60740478	
Project: Chowder Bay DFI		**Logged by:** AL	**Checked by:** PJW
Location: Chowder Bay		**Start Date:** 29/04/26	**End Date:** 30/04/26
			Location Meth.: dGPS0.1
Driller: Hard Access Drilling	**Hole Diameter:** 125-96 mm	**Easting:** 338457.54	**RL:** 26.1 m
Drill Rig: Man Portable Rig	**Inclination:** 90°	**Northing:** 6254468.30	**Ver. Datum:** AHD
Total Depth: 13.05 m	**Bearing:** N/A	**Hor. Datum:** MGA2020-56	**Surface:** Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	7		C: 10.57-10.66 m Is(50) D=1.3 MPa; C: 10.57-10.66 m Is(50) A=1.2 MPa; C: 11.57-11.65 m Is(50) D=0.47 MPa; C: 11.57-11.65 m Is(50) A=0.85 MPa	16.0, 15.0	10.00–11.78 (11.0)	SANDSTONE: fine to medium grained, pale grey mottled brown, cross-bedded at 0-20°, with dark grey laminations .	SW				10.00 m: DB; 10.77 m: P, 10°, RF, PR, CN; 11.00 m: DB; 11.09 m: DB; 11.28 m: DB; 11.65 m: P, 20°, RF, PR, CN; 11.72 m: P, 5°, RF, PR, SN X
	11.78 / 8		C: 12.41-12.50 m Is(50) D=1.3 MPa; C: 12.41-12.50 m Is(50) A=0.67 MPa	14.0	11.78–12.50 (12.0)	SANDSTONE: fine to medium grained, pale grey, indistinctly bedded, with black flecks.		100 (100) [100]			11.78 m: DB; 11.87 m: HB; 11.97 m: MB; 12.00 m: MB
				13.0	12.50–13.05 (13.0)	SANDSTONE: fine to medium grained, pale grey mottled black flecks, cross-bedded at 0-10° .					12.50 m: DB; 12.73 m: DB; 13.00 m: DB
				12.0	14.0, 15.0	Terminated at 13.05 m. Target depth.					

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

CORE PHOTO REPORT

PROJECT : Chowder Bay DFI
LOCATION : Chowder Bay

ID NO. : ABH1/MW008
PROJECT NO. : 60740478

1 - 0.30 to 5.00 m

2 - 5.00 to 10.00 m

3 - 10.00 to 13.05 m

AECOM **Engineering Log**

DRAFT

Borehole No. **ABH2/LDA**

Sheet 1 of 1

Client: Department of Defence		**Project No:** 60740478	
Project: Chowder Bay DFI		**Logged by:** AL	**Checked by:** PJW
Location: Chowder Bay		**Start Date:** 07/05/26	**End Date:** 07/05/26
			Location Meth.: dGPS0.1
Driller: Hard Access Drilling	**Hole Diameter:** 200 mm	**Easting:** 338501.40	**RL:** 19.878 m
Drill Rig: Man Portable Rig	**Inclination:** 90°	**Northing:** 6254458.52	**Ver. Datum:** AHD
Total Depth: 2 m	**Bearing:** N/A	**Hor. Datum:** MGA2020-56	**Surface:** Concrete

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
DT	U	CONCC: 0.00-0.13 m	Not Encountered		0.0			FILL - CONCRETE: pale grey, aggregates size up to 52mm, with visible voids.	D		CONCRETE LAYER
AS		B: 0.13-1.00 m		19.0				FILL - Clayey Gravelly SAND: fine to coarse grained, dark brown; gravel, fine to coarse, sub-angular to sub-rounded, sandstone and coal wash.	M		FILL
		B: 1.00-2.00 m		18.0	1.0			FILL - Silty Gravelly SAND: fine to coarse grained, yellow-brown; gravel, fine to medium, sub-rounded, sandstone.	D		
				17.0	2.0			Terminated at 2.00 m. Target depth.			
				16.0	3.0						
				15.0	4.0						
					5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DRAFT

AECOM Engineering Log

Borehole No. ABH2/MW009

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 06/05/26
End Date: 06/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 8.7 m
Hole Diameter: 96-100 mm
Inclination: 90°
Bearing: N/A
Easting: 338511.26
Northing: 6254456.34
Hor. Datum: MGA2020-56
RL: 19.878 m
Ver. Datum: AHD
Surface: Concrete

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
DT		CONCC: 0.00-0.12 m ES: 0.00-0.15 m			0.0			FILL - CONCRETE: pale grey, aggregates size up to 25mm, with visible voids.	D		CONCRETE LAYER
HA		D: 0.12-0.33 m						FILL - Gravelly SAND: fine to coarse grained, dark brown; gravel, fine to coarse, sub-rounded, sandstone.	M		FILL
AD/T	U	D: 0.33-1.00 m		19.0				FILL - Clayey SAND: fine to coarse grained, brown; trace gravel, fine to medium, sub-angular to sub-rounded, sandstone .	M		
		D: 1.00-2.00 m ES: 1.00-2.00 m		18.0	1.0			FILL - Silty Gravelly SAND: fine to coarse grained, yellow-brown mottled grey; gravel, fine to medium, sub-rounded, sandstone .	D		
		SPT: 2.00-2.45 m 1,1,1 N=2 D: 2.00-2.45 m D: 2.00-3.00 m		17.0	2.0				M		
		SPT: 3.50-3.95 m 3,3,2 N=5 D: 3.50-3.95 m		16.0	3.0						
		SPT: 4.00-4.45 m 8,3,7 N=10 ES: 4.00-4.45 m		15.0	4.0						
					5.0						

Log continued on next page.

AECOM Engineering Log

DRAFT

Cored Borehole No. ABH2/MW009
Sheet 2 of 3

Client: Department of Defence		**Project No:** 60740478	
Project: Chowder Bay DFI		**Logged by:** AL	**Checked by:** PJW
Location: Chowder Bay		**Start Date:** 06/05/26	**End Date:** 06/05/26
			Location Meth.: dGPS0.1
Driller: Hard Access Drilling	**Hole Diameter:** 96-100 mm	**Easting:** 338511.26	**RL:** 19.878 m
Drill Rig: Man Portable Rig	**Inclination:** 90°	**Northing:** 6254456.34	**Ver. Datum:** AHD
Total Depth: 8.7 m	**Bearing:** N/A	**Hor. Datum:** MGA2020-56	**Surface:** Concrete

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
				19.0, 18.0, 17.0, 16.0	0.0, 1.0, 2.0, 3.0, 4.0		*Log continued from previous page.*					
NMLC		08/05/2026 14:09:21					SANDSTONE: medium to coarse grained, yellow-brown mottled red, indistinctly bedded, trace dark brown laminations and iron staining .	HW	23 (13) [0]	●		HAWKESBURY SANDSTONE 4.56-4.67 m: CS, 5°, RF, PR, SN Fe, 110mm 4.73-4.77 m: CS, 5°, RF, PR, VN Clay, 40mm
			C: 4.72-4.77 m Is(50) A=0.26 MPa C: 4.72-4.77 m	15.0	5.0		NO CORE: from 4.77 to 5.68m.					

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AECOM Engineering Log

Cored Borehole No. ABH2/MW009

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 06/05/26
End Date: 06/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 8.7 m

Hole Diameter: 96-100 mm
Inclination: 90°
Bearing: N/A

Easting: 338511.26
Northing: 6254456.34
Hor. Datum: MGA2020-56
RL: 19.878 m
Ver. Datum: AHD
Surface: Concrete

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	1	▼				NO CORE: from 4.77 to 5.68m.			
	5.68 / 2		C: 5.76-5.89 m Is(50) D=0.39 MPa; C: 5.76-5.89 m Is(50) A=0.5 MPa	14.0	6.0	SANDSTONE: fine to coarse grained, brown mottled grey, cross-bedded at 0-30°.	MW	100 (100) [94]	5.87 m: DB 6.00 m: MB 6.07 m: P, 10°, RF, PR, SN Fe 6.10 m: P, 5°, RF, PR, SN Fe 6.15 m: P, 30°, RF, PR, SN Fe 6.16 m: J, 55°, RF, PR, SN Fe, 0.1 m persistence 6.28 m: P, 5°, RF, PR, SN Fe
			C: 6.72-6.82 m Is(50) D=0.69 MPa; C: 6.72-6.82 m Is(50) A=0.69 MPa	13.0		SANDSTONE: fine to medium grained, pale grey mottled brown, indistinctly bedded, with dark grey flecks and iron stained bands.	SW		6.33 m: DB 6.50 m: DB 6.52 m: P, 10°, RF, PR, CN 6.59 m: DB 6.72 m: DB 6.83 m: P, 5°, RF, PR, SN Fe 6.86 m: P, 20°, RF, PR, SN Fe
					7.0	SANDSTONE: fine to coarse grained, brown-grey, cross-bedded at 0-15°, with brown laminations and quartz inclusions, trace iron stained bands.			6.90 m: DB 7.00 m: MB 7.08 m: P, 20°, RF, PR, CN 7.22 m: P, 0°, RF, PR, SN Fe
			C: 7.54-7.65 m Is(50) D=0.4 MPa; C: 7.54-7.65 m Is(50) A=0.59 MPa	12.0		SANDSTONE: fine to medium grained, dark brown mottled grey, cross-bedded at 0-20°, with quartz inclusions and dark brown laminations.			7.44 m: DB 7.53 m: DB 7.72 m: P, 5°, RF, PR, CN 7.77 m: DB 7.92 m: P, 10°, RF, PR, CT Clay
			C: 8.28-8.37 m Is(50) D=0.59 MPa; C: 8.28-8.37 m Is(50) A=0.77 MPa		8.0	SANDSTONE: fine to coarse grained, orange-brown mottled grey, cross-bedded at 0-15°, with quartz inclusions and dark grey flecks.	MW		8.00 m: MB 8.09 m: P, 5°, RF, PR, CN 8.26 m: P, 10°, RF, UN, SN Fe 8.28 m: DB
						SANDSTONE: fine to medium grained, dark grey, indistinctly bedded, with quartz inclusions and iron stained bands.			8.44 m: DB 8.46 m: DB 8.57 m: P, 5°, RF, PR, CN 8.64 m: P, 5°, RF, PR, CN
				11.0	9.0	Terminated at 8.70 m. Target depth.			
				10.0	10.0				

INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ — SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH

DEFECT SPACING (mm) 20 60 200 600 2000

AECOM
PROJECT: CHOWDER BAY
PROJECT No.: 60740478
CLIENT: Department of Defence
BOREHOLE No.: ABH2/MW009
DEPTH: 4.50 - 8.70m
DATE: 6/5/2026
KODAK Gray Scale
KODAK Color Control Patches
METRE STICK
4m CORING STARTS AT 4.50m
4.77m NO CORE
5m NO CORE
5.68m
6m
7m
8m END OF HOLE 8.70m

1 - 4.50 to 8.70 m

DRAFT

Engineering Log

Borehole No. ABH3

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 30/04/26
End Date: 01/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Hand Auger
Total Depth: 20.88 m
Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A
Easting: 338486.75
Northing: 6254432.45
Hor. Datum: MGA2020-56
RL: 25.77 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	D: 0.00-1.00 m	Not Measured		0.0			TOPSOIL - Silty SAND: fine to medium grained, brown; trace gravel, fine to medium sub-rounded, sandstone .	D		TOPSOIL
		ES: 0.15-0.40 m PID=0.1 PPM						FILL - Sandy CLAY: low plasticity, pale brown mottled orange; sand, fine to medium grained; with gravel, fine to coarse, sub-angular to sub-rounded, sandstone; trace cobbles, sub-rounded, size up to 67mm, sandstone.	w < PL		FILL
		ES: 0.40-0.60 m PID=0.1 PPM						FILL - Silty Gravelly SAND: fine to medium grained, pale brown mottled orange; gravel, fine to coarse, sub-rounded, sandstone .	D		
AD/T				25.0							
		D: 1.00-2.00 m			1.0						
		SPT: 1.50-1.60 m 10/100 mm HB N=R D: 1.50-1.51 m									
				24.0							
		D: 2.00-3.00 m			2.0						
				23.0							
		SPT: 3.00-3.45 m 3,4,1 N=5 D: 3.00-3.45 m			3.0						
				22.0							
					4.0						
		SPT: 4.50-4.95 m 8,6,13 N=19 D: 4.50-4.95 m						FILL - Gravelly SAND: fine to coarse grained, yellow-brown; gravel, fine to coarse, sub-rounded, sandstone.			
				21.0							
					5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

AECOM **Engineering Log**

DRAFT

Borehole No. **ABH3**

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 30/04/26
End Date: 01/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Hand Auger
Total Depth: 20.88 m

Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A

Easting: 338486.75
Northing: 6254432.45
Hor. Datum: MGA2020-56

RL: 25.77 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	U		Not Measured					FILL - Gravelly SAND: fine to coarse grained, yellow-brown; gravel, fine to coarse, sub-rounded, sandstone.			
				20.0							
					6.0						
				19.0							
					7.0						
				18.0							
					8.0						
				17.0							
					9.0						
				16.0							
					10.0						

Log continued on next page.

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 30/04/26
End Date: 01/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Hand Auger
Total Depth: 20.88 m

Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A

Easting: 338486.75
Northing: 6254432.45
Hor. Datum: MGA2020-56

RL: 25.77 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ (SOIL, VL, L, M, H, VH, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					5.0		*Log continued from previous page.*					
NMLC	1	Not Measured	C: 5.55-5.64 m Is(50) D=0.91 MPa; C: 5.55-5.64 m Is(50) A=0.86 MPa	20.0	6.0		FILL - COBBLES/BOULDERS: medium to coarse grained, dark brown mottled orange; sandstone.		100 (0) [0]	●		FILL
	6.05						FILL - COBBLES/BOULDERS: dark grey; shale.		45 (0) [0]			
	2						FILL - COBBLES/BOULDERS: fine to coarse grained, red-brown, iron stained, with elongated carbonaceous fragments; sandstone.					
			C: 6.56-6.66 m Is(50) D=0.29 MPa; C: 6.56-6.66 m Is(50) A=0.46 MPa	19.0	7.0		FILL - COBBLES/BOULDERS: fine to coarse grained, pale brown, with elongated carbonaceous fragments; sandstone.			●o		
							FILL - COBBLES/BOULDERS: fine to coarse grained, red-brown, iron stained; sandstone.					
			C: 7.42-7.54 m Is(50) D=0.84 MPa; C: 7.42-7.54 m Is(50) A=0.91 MPa				FILL - COBBLES/BOULDERS: fine to coarse grained, pale grey mottled brown, with dark grey laminations; sandstone. from 7.15m: pale grey mottled brown			●		
				18.0, 17.0	8.0, 9.0		NO CORE: from 7.55 to 9.40m.					
	9.40						FILL - Sandy CLAY: medium plasticity, yellow-brown; sand, fine to medium grained; trace gravel, fine to coarse, sub-angular to sub-rounded, sandstone.		37 (0) [0]			
			C: 9.67-9.77 m Is(50) D=0.63 MPa; C: 9.67-9.77 m Is(50) A=0.69 MPa	16.0	10.0		FILL - COBBLES/BOULDERS: fine to medium grained, pale grey, trace carbonaceous flecks; sandstone.			●		

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AECOM **Engineering Log**

Cored Borehole No. ABH3

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 30/04/26
End Date: 01/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Hand Auger
Total Depth: 20.88 m

Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A

Easting: 338486.75
Northing: 6254432.45
Hor. Datum: MGA2020-56

RL: 25.77 m
Ver. Datum: AHD
Surface: Topsoil

Column headings: METHOD | CORE RUN | GROUND WATER | SAMPLES & FIELD TESTS | RL (m AHD) | DEPTH (m) | GRAPHIC LOG | MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density) | WEATHERING | TCR (SCR) [RQD] (%) | INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH) | DEFECT SPACING (mm) (20, 60, 200, 600, 2000) | ADDITIONAL OBSERVATIONS (Defect Descriptions)

Method: NMLC. Core runs: 3 (to 12.40), 4 (from 12.40). Ground water: Not Measured.

RL scale: 15.0 at 10.33 m depth; 14.0 at 11.33 m; 13.0 at 12.77 m; 12.0 at 13.77 m; 11.0 at 14.77 m.

Depth (m)	Samples & Field Tests	Material Description	TCR (SCR) [RQD]	Additional Observations
10.00–10.50	C: 10.08-10.15 m Is(50) D=0.41 MPa; C: 10.08-10.15 m Is(50) A=0.45 MPa	FILL - COBBLES/BOULDERS: fine to medium grained, pale grey, trace carbonaceous flecks; sandstone. from 10.00m: orange-brown		
10.50–12.40		NO CORE: from 10.50 to 12.40m.		
12.40		FILL - COBBLES/BOULDERS: fine to coarse grained, brown-grey; sandstone.	61 (61) [41]	12.40 m: P, 10°, RF, PR, CN 12.45 m: J, 65°, RF, PR, SN Fe
	PP=160,180,170 kPa	FILL - Silty CLAY: low plasticity, yellow-grey.		12.50 m: P, 25°, RF, PR, CT Clay
	C: 12.76-12.85 m Is(50) D=0.27 MPa; C: 12.76-12.85 m Is(50) A=0.35 MPa	FILL - COBBLES/BOULDERS: fine to coarse grained, pale brown mottled yellow, iron stained trace brown laminations; sandstone.		12.73 m: P, 15°, RF, PR, SN Fe 12.78 m: DB 13.00 m: DB
		FILL - COBBLES/BOULDERS: fine to coarse grained, brown-grey, with dark brown laminations; sandstone.		13.13 m: P, 5°, RF, PR, SN Fe 13.29 m: P, 15°, RF, PR, SN Fe 13.34 m: DB
		FILL - COBBLES/BOULDERS: fine to coarse grained, red-brown mottled grey; sandstone.		13.41 m: DB 13.48 m: DB
	C: 13.58-13.67 m Is(50) D=0.27 MPa; C: 13.58-13.67 m Is(50) A=0.82 MPa	FILL - COBBLES/BOULDERS: fine to medium grained, pale brown-grey, with dark grey laminations and flecks; sandstone.		13.59 m: P, 5°, RF, PR, CN 13.89 m: P, 5°, RF, PR, VN Clay
14.00		FILL - COBBLES/BOULDERS: fine to coarse grained, brown, iron stained, with brown laminations; sandstone.		14.00 m: DB 14.03 m: DB 14.04 m: P, 15°, RF, UN, SN Fe 14.18 m: P, 5°, RF, PR, CN 14.28 m: P, 10°, RF, PR, CN
	C: 14.35-14.37 m Is(50) D=0.59 MPa; C: 14.35-14.37 m Is(50) A=0.61 MPa	FILL - COBBLES/BOULDERS: fine to coarse grained, red-brown; sandstone.		14.34-14.36 m: EW, 10°, RF, PR, VN Clay, 60mm 14.39 m: P, 5°, RF, PR, CN 14.46 m: P, 15°, RF, PR, SN Fe 14.50 m: P, 15°, RF, PR, SN Fe

AECOM **Engineering Log**

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Cored Borehole No. ABH3

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 30/04/26
End Date: 01/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Hand Auger
Total Depth: 20.88 m
Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A
Easting: 338486.75
Northing: 6254432.45
Hor. Datum: MGA2020-56
RL: 25.77 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	15.90	Not Measured		10.0	15.0	FILL - COBBLES/BOULDERS: fine to coarse grained, red-brown; sandstone.			15.69 m: DB
	5		C: 16.66-16.74 m Is(50) D=0.86 MPa C: 16.66-16.74 m Is(50) A=0.73 MPa	9.0	16.0	SANDSTONE: fine to medium grained, yellow-brown mottled red, cross-bedded at 0-25°, with iron stained.	MW	55 (48) [20]	HAWKESBURY SANDSTONE 15.94 m: DB 16.00 m: DB 16.28 m: P, 10°, RF, PR, CN 16.37 m: P, 10°, RF, UN, CN 16.50 m: P, 5°, RF, PR, CN 16.58 m: P, 20°, RF, PR, CN 16.65 m: P, 15°, RF, IR, CN 16.73 m: P, 15°, RF, PR, CN 16.79 m: P, 10°, RF, PR, CN 16.85 m: P, 20°, RF, PR, CN 16.89 m: J, 75°, RF, PR, SN Fe 16.90 m: P, 15°, RF, PR, SN Fe 16.93-17.00 m: CS
	17.90			8.0	17.0	NO CORE: from 17.0 to 17.90m.			
	6		C: 17.90-17.96 m Is(50) D=0.71 MPa C: 17.90-17.96 m Is(50) A=0.87 MPa C: 18.66-18.76 m Is(50) D=0.66 MPa C: 18.66-18.76 m Is(50) A=0.89 MPa C: 19.69-19.77 m Is(50) D=0.77 MPa C: 19.69-19.77 m	7.0 6.0	18.0 19.0 20.0	SANDSTONE: fine to medium grained, pale grey, indistinctly bedded, trace black laminations.	FR	89 (89) [85]	17.96 m: P, 30°, RF, PR, CT X 18.00 m: P, 5°, RF, PR, CN 18.24 m: DB 19.00 m: DB 19.10 m: P, 20°, RF, PR, CN

Inferred strength columns: SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH; 20 x Is(50) A: ● D: o; UCS: □. Defect spacing (mm): 20, 60, 200, 600, 2000.

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Engineering Log

Cored Borehole No. ABH3

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 30/04/26
End Date: 01/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Hand Auger
Total Depth: 20.88 m
Hole Diameter: 125-96 mm
Inclination: 90°
Bearing: N/A
Easting: 338486.75
Northing: 6254432.45
Hor. Datum: MGA2020-56
RL: 25.77 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x $Is_{(50)}$ A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC		Not Measured	Is(50) A=0.74 MPa C: 19.78-20.00 m UCS=23 MPa C: 20.35-20.45 m Is(50) D=0.59 MPa C: 20.35-20.45 m Is(50) A=0.69 MPa	5.0	20.0		SANDSTONE: fine to medium grained, pale grey, indistinctly bedded, trace black laminations.	FR				20.00 m: DB 20.35 m: P, 5°, RF, PR, CN 20.46 m: DB
				4.0, 3.0, 2.0, 1.0	21.0, 22.0, 23.0, 24.0, 25.0		Terminated at 20.88 m. Target depth.					

CORE PHOTO REPORT

PROJECT : Chowder Bay DFI
LOCATION : Chowder Bay

PROJECT NO. : 60740478

1 - 5.50 to 10.00 m

2 - 10.00 to 15.00 m

3 - 15.00 to 20.00 m

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4 - 20.00 to 20.88 m

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AECOM **Engineering Log**

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Borehole No. **ABH4**

Client: Department of Defence		**Project No:** 60740478	
Project: Chowder Bay DFI		**Logged by:** AL	**Checked by:** PJW
Location: Chowder Bay		**Start Date:** 04/05/26	**End Date:** 04/05/26
			Location Meth.: dGPS0.1
Driller: Hard Access Drilling	**Hole Diameter:** 96-125 mm	**Easting:** 338509.27	**RL:** 25.743 m
Drill Rig: Jacro200	**Inclination:** 90°	**Northing:** 6254423.61	**Ver. Datum:** AHD
Total Depth: 18.56 m	**Bearing:** N/A	**Hor. Datum:** MGA2020-56	**Surface:** Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	B: 0.00-1.00 m	Not Measured		0.0			TOPSOIL - Clayey SAND: fine to coarse grained, dark brown; trace rootlets .	D		TOPSOIL
				25.0				FILL - Clayey SAND: fine to coarse grained, brown; trace gravel, fine to coarse, sub-rounded, sandstone .			FILL
		B: 1.00-1.50 m ES: 1.00-1.50 m PID=0.1 PPM			1.0						
AD/T		SPT: 1.50-1.95 m 11,8,5 N=13 B: 1.50-2.00 m D: 1.50-1.95 m		24.0				FILL - Silty Gravelly SAND: fine to medium grained, pale brown-grey mottled yellow; gravel, fine to coarse, sub-rounded, sandstone .			
					2.0						
				23.0							
		SPT: 3.00-3.40 m 3,10,14/100 mm HB N=R D: 3.00-3.40 m			3.0						
				22.0					D		
					4.0						
		SPT: 4.20-4.65 m 6,6,7 N=13 D: 4.20-4.65 m						FILL - Gravelly Silty SAND: fine to medium grained, pale brown-grey; gravel, fine to coarse, sub-rounded, sandstone ; with clay.			
				21.0							
					5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

AECOM **Engineering Log**

DRAFT

Borehole No. **ABH4**

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 04/05/26
End Date: 04/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Jacro200
Total Depth: 18.56 m

Hole Diameter: 96-125 mm
Inclination: 90°
Bearing: N/A

Easting: 338509.27
Northing: 6254423.61
Hor. Datum: MGA2020-56

RL: 25.743 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	U	SPT: 6.00-6.45 m 5,9,14 N=23 D: 6.00-6.45 m	Not Measured	20.0, 19.0	6.0, 7.0			FILL - Gravelly Silty SAND: fine to medium grained, pale brown-grey; gravel, fine to coarse, sub-rounded, sandstone ; with clay.			
		SPT: 7.30-7.37 m 8/70 mm HB N=R D: 7.30-7.37 m		18.0, 17.0, 16.0	8.0, 9.0, 10.0			*Log continued on next page.*			

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

AECOM Engineering Log

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 04/05/26
End Date: 04/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Jacro200
Total Depth: 18.56 m

Hole Diameter: 96-125 mm
Inclination: 90°
Bearing: N/A

Easting: 338509.27
Northing: 6254423.61
Hor. Datum: MGA2020-56

RL: 25.743 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
				20.0, 19.0	5.0, 6.0, 7.0		*Log continued from previous page.*					
NMLC	1	Not Measured		18.0			FILL - COBBLES/BOULDERS: dark grey; shale.		26 (0) [0]			FILL 7.37 m: P, 35°, RF, PR, CN
			C: 7.60-7.68 m Is(50) D=0.78 MPa C: 7.60-7.68 m Is(50) A=0.98 MPa				FILL - COBBLES/BOULDERS: fine to medium grained, dark grey; sandstone.					7.68 m: P, 5°, RF, PR, SN Fe
			C: 8.25-8.29 m Is(50) A=0.2 MPa C: 8.25-8.29 m		8.0		FILL - COBBLES/BOULDERS: fine to coarse grained, brown-grey mottled red; sandstone.					7.89-8.03 m: P, 0-15°, RF, PR, SN Fe, 30 mm spacing, x 5 8.10 m: J, 90°, RF, PR, SN Fe, 100 mm persistence 8.15-8.24 m: DB, RF 8.30 m: P, 5°, RF, PR, SN Fe 8.34 m: J, 80°, RF, PR, SN Fe, 110 mm persistence 8.35 m: P, 15°, RF, IR, SN Fe 8.42 m: DB
				17.0, 16.0	9.0, 10.0		NO CORE: from 8.42 to 11.60m.					

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Engineering Log

Cored Borehole No. ABH4

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 04/05/26
End Date: 04/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Jacro200
Total Depth: 18.56 m

Hole Diameter: 96-125 mm
Inclination: 90°
Bearing: N/A

Easting: 338509.27
Northing: 6254423.61
Hor. Datum: MGA2020-56

RL: 25.743 m
Ver. Datum: AHD
Surface: Topsoil

Method: NMLC. Core run: 2 (from 11.60). Ground water: Not Measured.

RL (m AHD) scale: 15.0, 14.0, 13.0, 12.0, 11.0. Depth (m) scale: 10.0, 11.0, 12.0, 13.0, 14.0, 15.0.

Inferred strength: 20 x Is(50), A: ● D: o, UCS: □; SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH. Defect spacing (mm): 20, 60, 200, 600, 2000.

DEPTH (m)	SAMPLES & FIELD TESTS	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	TCR (SCR) [RQD] (%)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
10.00–11.60		NO CORE: from 8.42 to 11.60m.		
11.60–12.00	C: 11.78-11.84 m Is(50) D=0.39 MPa; C: 11.78-11.84 m Is(50) A=0.47 MPa	FILL - COBBLES/BOULDERS: fine to medium grained, pale grey mottled yellow and red, iron stained; sandstone.	40 (0) [0]	11.60-11.66 m: CS, RF, IR, SN Fe, 60 mm thick 11.67 m: J, 55°, RF, PR, SN Fe, 50 mm persistence 11.76 m: P, 5°, RF, PR, CN 11.78 m: P, 10°, RF, PR, CN 11.84-12.40 m: P, 5-40°, RF, PR, SN Fe, 30-250 mm spacing, x 6
12.00–12.26		FILL - COBBLES/BOULDERS: fine to medium grained, pale grey; sandstone.		
12.26–12.81	C: 12.42-12.50 m Is(50) D=0.17 MPa; C: 12.42-12.50 m Is(50) A=0.18 MPa; PP=50,80,80 kPa	FILL - COBBLES/BOULDERS: fine to coarse grained, red-brown mottled yellow; sandstone.		12.63 m: P, 25°, RF, PR, CN, healed 12.71 m: P, 15°, RF, PR, SN Fe 12.75 m: P, 5°, RF, PR, CN
12.81–12.87		FILL - Gravelly CLAY: medium plasticity, brown mottled red; gravel, medium to coarse, sub-angular, sandstone.		12.81-12.87 m: EW, 5°, RF, PR, VN Clay, 60 mm thick 12.90-13.00 m: MB, x 3 13.00-13.09 m: CS, RF, IR, VN Clay, 90 mm thick
12.87–13.21	C: 13.17-13.21 m Is(50) A=0.4 MPa; C: 13.17-13.21 m	FILL - COBBLES/BOULDERS: fine to coarse grained, red-brown, iron stained; sandstone.		13.12 m: P, 5°, RF, PR, CN, healed 13.16 m: P, 5°, RF, PR, SN Fe 13.21 m: DB
13.21–15.00		NO CORE: from 13.21 to 15.60m.		

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AECOM **Engineering Log**

Cored Borehole No. ABH4

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL **Checked by:** PJW
Start Date: 04/05/26 **End Date:** 04/05/26
Location Meth.: dGPS0.1

Driller: Hard Access Drilling
Drill Rig: Jacro200
Total Depth: 18.56 m

Hole Diameter: 96-125 mm
Inclination: 90°
Bearing: N/A

Easting: 338509.27 **RL:** 25.743 m
Northing: 6254423.61 **Ver. Datum:** AHD
Hor. Datum: MGA2020-56 **Surface:** Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	DEPTH (m)	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC	3 (15.60)	Not Measured		15.00–15.60	NO CORE: from 13.21 to 15.60m.			
			C: 15.69-15.76 m Is(50) D=0.11 MPa; C: 15.69-15.76 m Is(50) A=0.13 MPa	15.60–16.00	SANDSTONE: fine to coarse grained, red-brown, indistinctly bedded.	HW	100 (96) [87]	15.60-15.63 m: CS, RF, IR, SN Fe, 30 mm thick HAWKESBURY SANDSTONE
				16.00–16.13	SANDSTONE: fine to coarse grained, pale grey mottled brown, indistinctly bedded, trace iron staining.	MW		16.00-16.10 m: P, 15-20°, RF, PR, CN, 30 spacing, X3
			C: 16.62-16.71 m Is(50) D=0.36 MPa; C: 16.62-16.71 m Is(50) A=0.39 MPa; C: 17.34-17.44 m Is(50) D=1.1 MPa; C: 17.34-17.44 m Is(50) A=1.9 MPa	16.13–17.37	SANDSTONE: fine to coarse grained, brown-grey, cross-bedded at 0-10°, 5 mm spacing, trace carbonaceous flecks and iron staining.	MW / HW		16.95 m: P, 10°, RF, PR, VN Clay 17.00 m: DB 17.13 m: P, 10°, RF, PR, SN Fe 17.30-17.37 m: CS, RF, PR, SN Fe, 70 mm thick
			C: 18.05-18.30 m UCS=32 MPa; C: 18.32-18.40 m Is(50) D=0.76 MPa; C: 18.32-18.40 m Is(50) A=1.2 MPa	17.37–18.56	SANDSTONE: fine to medium grained, pale brown, cross-bedded at 0-30°, 20-90 mm spacing, trace carbonaceous laminations.	SW		17.85-18.05 m: MB, x 3
				18.56	Terminated at 18.56 m. Target depth.			

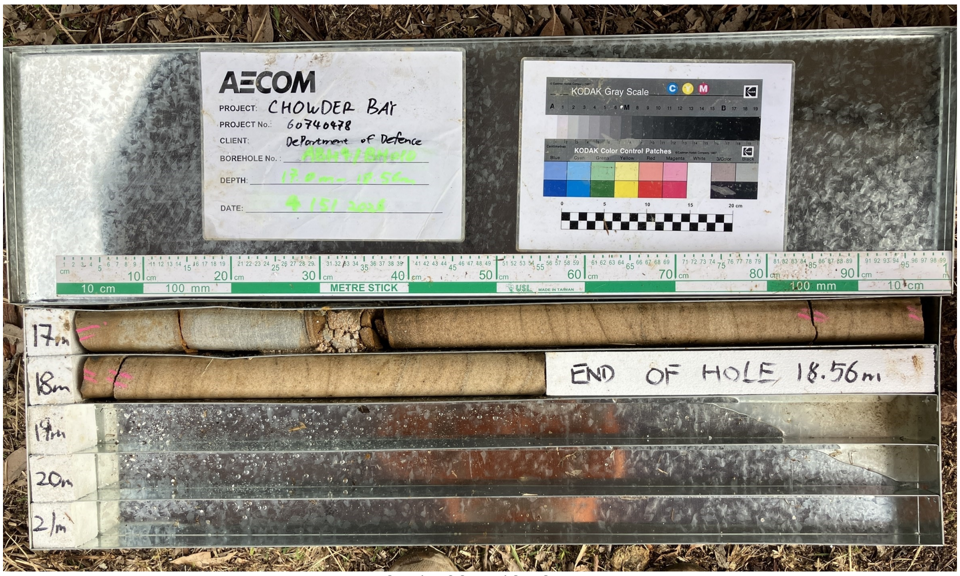
RL (m AHD): 10.0, 9.0, 8.0, 7.0, 6.0
DEPTH (m): 15.0, 16.0, 17.0, 18.0, 19.0, 20.0

INFERRED STRENGTH 20 x $Is_{(50)}$ A: ● D: o UCS: □ — SOIL, VL, L, M, H, VH, EH (0-6, 2, 6, 20, 60, 200)

DEFECT SPACING (mm): 20, 60, 200, 600, 2000

1 - 7.30 to 12.00 m

2 - 12.00 to 17.00 m



3 - 17.00 to 18.56 m

DRAFT

Engineering Log

Borehole No. ABH5

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: JL
Start Date: 07/05/26
Checked by: PJW
End Date: 07/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 8.7 m
Hole Diameter: 125-96-200 mm
Inclination: 90°
Bearing: N/A
Easting: 338531.07
Northing: 6254423.18
Hor. Datum: MGA2020-56
RL: 19.976 m
Ver. Datum: AHD
Surface: Concrete

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
DT	U		Not Encountered		0.0			FILL - CONCRETE: grey, aggregates up to 20 mm, 125 mm thick, road surface.	D		CONCRETE LAYER
AS		B: 0.13-1.50 m		19.0	1.0			FILL - Sandy Silty CLAY: low to medium plasticity, brown and dark grey; sand, fine to coarse grained; trace gravel, fine to coarse, sub-angular, sandstone.	M		FILL
AD/T		SPT: 1.50-1.95 m 5,3,0 N=3 D: 1.50-1.80 m; D: 2.00-3.00 m; SPT: 3.00-3.45 m 2,1,3 N=4 D: 3.00-3.45 m PP=200 kPa; PP=150 kPa; PP=200 kPa D: 3.50-4.50 m		18.0; 17.0	2.0; 3.0			FILL - Sandy CLAY: low plasticity, red-brown and dark grey; sand, fine to coarse grained.	w > PL		at 1.70 m: boulder encountered
		SPT: 4.50-4.55 m 25/50 mm HB N=R D: 4.50-4.55 m		16.0; 15.0	4.0; 5.0		CL	Sandy CLAY: low plasticity, red-brown; sand, fine to coarse grained .	M	H	EXTREMELY WEATHERED MATERIAL

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

AECOM Engineering Log

DRAFT

Borehole No. ABH5

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: JL
Start Date: 07/05/26
Checked by: PJW
End Date: 07/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 8.7 m

Hole Diameter: 125-96-200 mm
Inclination: 90°
Bearing: N/A

Easting: 338531.07
Northing: 6254423.18
Hor. Datum: MGA2020-56

RL: 19.976 m
Ver. Datum: AHD
Surface: Concrete

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
AD/T	U		Not Encountered				CL	Sandy CLAY: low plasticity, red-brown; sand, fine to coarse grained .		H	
								Log continued on next page.			
				14.0	6.0						
				13.0	7.0						
				12.0	8.0						
				11.0	9.0						
				10.0	10.0						

AECOM Engineering Log

Cored Borehole No. ABH5

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: JL
Start Date: 07/05/26
Checked by: PJW
End Date: 07/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 8.7 m

Hole Diameter: 125-96-200 mm
Inclination: 90°
Bearing: N/A

Easting: 338531.07
Northing: 6254423.18
Hor. Datum: MGA2020-56
RL: 19.976 m
Ver. Datum: AHD
Surface: Concrete

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					5.0	*Log continued from previous page.*			
NMLC	1	Not Encountered	C: 5.90-6.00 m Is(50) D=0.019 MPa; C: 5.90-6.00 m Is(50) A=0.011 MPa	14.0	6.0	Sandy CLAY: low plasticity, red-brown; sand, fine to coarse grained.	XW	98 (94) [22]	EXTREMELY WEATHERED MATERIAL
			C: 6.46-6.56 m Is(50) D=0.59 MPa; C: 6.46-6.56 m Is(50) A=1.1 MPa	13.0	7.0	SANDSTONE: fine to medium grained, pale grey and red-brown, bedded at 0-15°, 10-100 mm spacing, trace quartz inclusions, iron stained.	HW / XW / HW / MW		HAWKESBURY SANDSTONE 6.47 m: J, 30°, RF, ST, SN Fe, D 6.54 m: P, 5°, RF, CU, SN Fe 6.62-6.70 m: EW, 10°, VR, IR, CT Fe, 80 mm thick 6.72 m: P, 5°, RF, PR, SN Fe 6.74-6.88 m: HB, x 2 7.33 m: P, 0°, RF, PR, VN Clay 7.46 m: P, 0°, RF, CU, SN Fe
			C: 7.46-7.57 m Is(50) D=0.42 MPa; C: 7.46-7.57 m Is(50) A=0.35 MPa	12.0	8.0				7.73 m: P, 10°, RF, PR, VN Clay 7.76 m: P, 5°, RF, PR, VN Clay 7.82-8.51 m: HB, x 3
			C: 8.08-8.29 m UCS=18 MPa; C: 8.28-8.36 m Is(50) D=0.49 MPa; C: 8.28-8.36 m Is(50) A=0.76 MPa			from 8.00 to 8.70 m: bedded at 0-15°, 10-90 mm spacing			
				11.0	9.0	Terminated at 8.70 m. Target depth.			
				10.0	10.0				

INFERRED STRENGTH (20 x Is(50); A: ● D: o; UCS: □): SOIL, VL, L, M, H, VH, EH (0-6, 2, 6, 20, 60, 200)

DEFECT SPACING (mm): 20, 60, 200, 600, 2000

CORE PHOTO REPORT

ID NO. : ABH5

PROJECT : Chowder Bay DFI
LOCATION : Chowder Bay

PROJECT NO. : 60740478
Sheet 1 of 1

1 - 5.70 to 8.70 m

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AECOM Engineering Log

Borehole No. ABH6/MW010

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay
Project No: 60740478
Logged by: JL
Checked by: PJW
Start Date: 08/05/26
End Date: 08/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 5.88 m
Hole Diameter: 125 mm
Inclination: 90°
Bearing: N/A
Easting: 338571.37
Northing: 6254415.13
Hor. Datum: MGA2020-56
RL: 27.311 m
Ver. Datum: AHD
Surface: Concrete

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
DT					0.0			FILL - CONCRETE: dark grey, aggregates up to 10 mm diameter, 60 mm thick; road surface.	D		CONCRETE LAYER
AD/T	U	D: 0.06-0.50 m		27.0				FILL - Gravelly SAND: fine to coarse grained, dark grey; gravel, fine to medium, sub-angular, sandstone; trace clay.	M		FILL
		D: 0.50-1.50 m SPT: 1.46-1.50 m 25/40 mm HB N=R		26.0	1.0		CL	Sandy CLAY: low plasticity, pale grey and brown; sand, fine to coarse grained.	M	H	EXTREMELY WEATHERED MATERIAL

Log continued on next page.

RL (m AHD)	DEPTH (m)
25.0	2.0
24.0	3.0
23.0	4.0
	5.0

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AECOM Engineering Log

Cored Borehole No. ABH6/MW010

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: JL
Checked by: PJW
Start Date: 08/05/26
End Date: 08/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 5.88 m
Hole Diameter: 125 mm
Inclination: 90°
Bearing: N/A
Easting: 338571.37
Northing: 6254415.13
Hor. Datum: MGA2020-56
RL: 27.311 m
Ver. Datum: AHD
Surface: Concrete

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
					0.0					
				27.0						
					1.0					
				26.0			*Log continued from previous page.*			
NMLC	1		C: 1.64-1.73 m Is(50) D=0.094 MPa; C: 1.64-1.73 m Is(50) A=0.11 MPa				SANDSTONE: fine to medium grained, pale grey, indistinctly bedded, with carbonaceous laminations and flecks, trace iron staining , trace quartz inclusions.	XW; HW	100 (100) [89]	HAWKESBURY SANDSTONE 1.58 m: P, 0°, RF, ST, VN Clay 1.64 m: P, 15°, RF, IR, SN Fe 1.91 m: HB
					2.0					2.13-2.14 m: EW, 15°, RF, PR, VN Clay, 10 mm thick
			C: 2.35-2.44 m Is(50) D=0.074 MPa; C: 2.35-2.44 m Is(50) A=0.18 MPa	25.0						2.34 m: P, 20°, RF, PR, VN Clay 2.51 m: P, 15°, RF, IR, VN Clay
	2.90						from 2.74 to 3.00 m: bedded at 0-5°, variable spacing		51 (22) [20]	2.90-3.08 m: DB
	2		C: 3.24-3.32 m Is(50) D=0.21 MPa; C: 3.24-3.32 m Is(50) A=0.19 MPa	24.0	3.0			MW; HW		3.32 m: P, 5°, RF, IR, VN Fe 3.44 m: DB 3.49 m: DB 3.52 m: DB 3.57 m: P, 5°, RF, IR, VN Fe 3.70 m: J, 90°, RF, UN, VN Fe, 900 mm persistence
				23.0	4.0					4.17 m: HB 4.21-4.23 m: EW, 5°, RF, PR, VN Clay, 20 mm thick 4.32 m: J, 70°, RF, PR, VN Fe, 70 mm persistence 4.44 m: P, 0°, RF, PR, CT Qtz
			C: 4.45-4.55 m Is(50) D=0.13 MPa; C: 4.45-4.55 m Is(50) A=0.11 MPa							4.80-4.89 m: EW, 5°, RF, IR, VN Fe, 90 mm thick
					5.0					

Inferred strength columns: SOIL, VL, L, M, H, VH, EH (20 x Is(50), A: ● D: o, UCS). Defect spacing (mm): 20, 60, 200, 600, 2000.

DRAFT

Engineering Log

Cored Borehole No. ABH6/MW010

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: JL
Checked by: PJW
Start Date: 08/05/26
End Date: 08/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 5.88 m
Hole Diameter: 125 mm
Inclination: 90°
Bearing: N/A
Easting: 338571.37
Northing: 6254415.13
Hor. Datum: MGA2020-56
RL: 27.311 m
Ver. Datum: AHD
Surface: Concrete

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □	DEFECT SPACING (mm)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC			C: 5.50-5.58 m Is(50) D=0.092 MPa; C: 5.50-5.58 m Is(50) A=0.083 MPa	22.0			SANDSTONE: fine to medium grained, pale grey, indistinctly bedded, with carbonaceous laminations and flecks, trace iron staining , trace quartz inclusions.	HW				5.22 m: P, 5°, RF, IR, VN Fe 5.29 m: P, 30°, RF, ST, VN Fe 5.33 m: P, 10°, RF, IR, VN Fe 5.37 m: P, 15°, RF, ST, VN Fe 5.60-5.67 m: EW, 5°, RF, IR, VN Clay and Fe, 70 mm thick
				21.0, 20.0, 19.0, 18.0	6.0, 7.0, 8.0, 9.0, 10.0		Terminated at 5.88 m. Target depth.					

Inferred strength scale: SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH
Defect spacing scale: 20, 60, 200, 600, 2000

1 - 1.50 to 5.88 m

DRAFT

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Engineering Log

Borehole No. ABH7

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 05/05/26
End Date: 05/05/26
Location Meth.: Handheld GPS

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 5.53 m

Hole Diameter: 96-125 mm
Inclination: 90°
Bearing: N/A

Easting: 322581.43
Northing: 13740838.65
Hor. Datum: MGA2020-56

RL: 37.111 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	B: 0.00-0.20 m	Not Measured	37.0	0.0			TOPSOIL - Silty SAND: fine to medium grained, dark brown; trace rootlets.	D		TOPSOIL
				36.0	1.0		CL	Sandy CLAY: red-brown; sand, fine to medium grained.	D	H	EXTREMELY WEATHERED MATERIAL
				35.0	2.0			*Log continued on next page.*			
				34.0	3.0						
				33.0	4.0						
					5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

AECOM **Engineering Log**

DRAFT

Cored Borehole No. ABH7

Client: Department of Defence		**Project No:** 60740478	
Project: Chowder Bay DFI		**Logged by:** AL	**Checked by:** PJW
Location: Chowder Bay		**Start Date:** 05/05/26	**End Date:** 05/05/26
			Location Meth.: Handheld GPS
Driller: Hard Access Drilling	**Hole Diameter:** 96-125 mm	**Easting:** 322581.43	**RL:** 37.111 m
Drill Rig: Man Portable Rig	**Inclination:** 90°	**Northing:** 13740838.65	**Ver. Datum:** AHD
Total Depth: 5.53 m	**Bearing:** N/A	**Hor. Datum:** MGA2020-56	**Surface:** Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
				37.0	0.0				
				36.0	1.0	*Log continued from previous page.*			
NMLC	1	Not Measured	C: 1.42-1.51 m Is(50) D=0.11 MPa; C: 1.42-1.51 m Is(50) A=0.22 MPa; PP=100,140,100 kPa			SANDSTONE: medium to coarse grained, yellow-brown mottled grey, cross-bedded at 0-15°, 5-100 mm spacing, trace carbonaceous laminations.	HW; MW; XW	100 (83) [67]	HAWKESBURY SANDSTONE 1.21 m: P, 15°, RF, IR, CN 1.27 m: P, 10°, RF, PR, SN Fe 1.51 m: P, 5°, RF, PR, CT Clay 1.71 m: P, 20°, RF, PR, CN 1.86-1.93 m: EW, 0-15°, RF, PR, VN Clay, 70 mm thick
				35.0	2.0	SANDSTONE: medium to coarse grained, brown grey, cross bedded at 0-20°, 10-90 mm spacing, trace carbonaceous laminations and flecks.	MW		2.00-2.33 m: MB, x 5
	2.33								
	2		C: 2.52-2.63 m Is(50) D=0.67 MPa; C: 2.52-2.63 m Is(50) A=0.53 MPa			SANDSTONE: fine to medium grained, pale grey mottled brown, cross-bedded at 0-15°, 5-40 mm spacing, with carbonaceous laminations, trace iron bands.	SW	100 (100) [86]	2.60 m: P, 10°, RF, PR, SN X 2.90 m: P, 5°, RF, PR, SN Fe 3.00 m: MB
			PP=40, 50, 40 kPa; C: 3.46-3.56 m Is(50) D=0.14 MPa; C: 3.46-3.56 m Is(50) A=0.28 MPa	34.0	3.0	SANDSTONE: medium to coarse grained, pale brown mottled grey and red, cross-bedded at 0-20°, 10-80 mm spacing, with carbonaceous laminations. from 3.15-3.60 m: iron stained bands	MW		3.13 m: P, 10°, RF, PR, SN X 3.19 m: P, 15°, RF, PR, SN X 3.28 m: P, 5°, RF, PR, VN Clay, 20 mm aperture 3.36 m: P, 10°, RF, CU, SN Fe 3.47 m: P, 20°, RF, PR, SN Fe 3.51 m: P, 0°, RF, PR, SN Fe
						SANDSTONE: fine to coarse grained, pale grey, cross bedded at 0-20°, 10-100 mm spacing, with carbonaceous laminations.	SW		3.74 m: P, 5°, RF, PR, CN
	3.90								3.90-4.02 m: MB, x 3
	3		C: 4.34-4.42 m Is(50) D=0.52 MPa; C: 4.34-4.42 m Is(50) A=0.84 MPa	33.0	4.0			100 (100) [97]	4.08-4.13 m: P, 5-10°, RF, PR, CN, 50 mm spacing, x 2 4.26-4.31 m: DB, x 2 4.33-4.64 m: P, 5-10°, RF, PR, CN, 60-250 mm spacing, x 3 4.80-5.00 m: MB, x 4
					5.0				

Graphic log columns: INFERRED STRENGTH (20 x $Is_{(50)}$; A: ● D: o; UCS: □) — SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH; DEFECT SPACING (mm) — 20, 60, 200, 600, 2000.

AECOM Engineering Log

DRAFT

Cored Borehole No. ABH7

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 05/05/26
End Date: 05/05/26
Location Meth.: Handheld GPS

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 5.53 m

Hole Diameter: 96-125 mm
Inclination: 90°
Bearing: N/A

Easting: 322581.43
Northing: 13740838.65
Hor. Datum: MGA2020-56

RL: 37.111 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC		Not Measured	C: 5.39-5.47 m Is(50) D=0.46 MPa; C: 5.39-5.47 m Is(50) A=0.76 MPa	32.0			SANDSTONE: fine to coarse grained, pale grey, cross bedded at 0-20°, 10-100 mm spacing, with carbonaceous laminations.	SW				5.32 m: DB 5.40 m: DB
				31.0, 30.0, 29.0, 28.0	6.0, 7.0, 8.0, 9.0, 10.0		Terminated at 5.53 m. Target depth.					

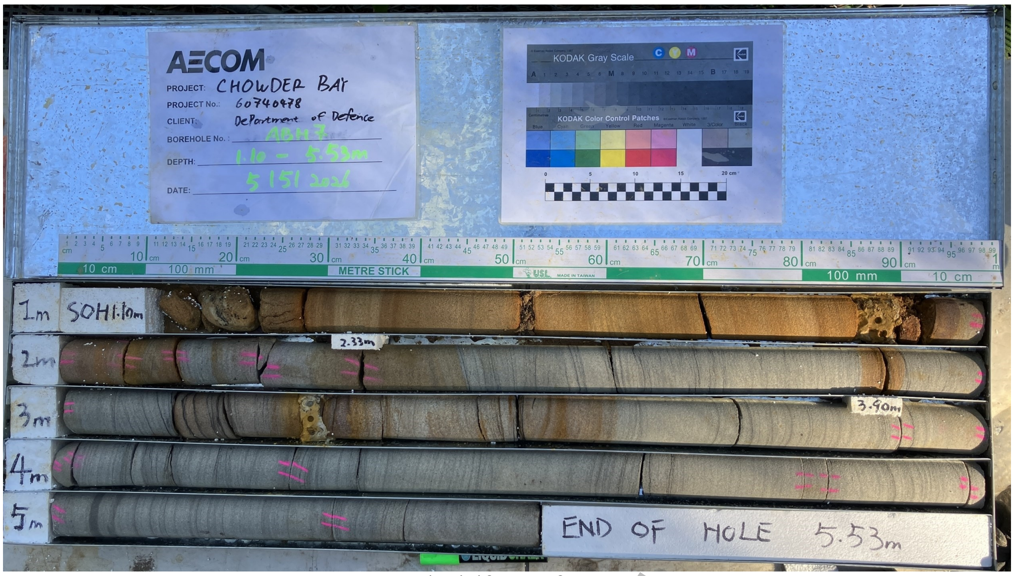
Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

CORE PHOTO REPORT

ID NO. : ABH7

PROJECT : Chowder Bay DFI
LOCATION : Chowder Bay

PROJECT NO. : 60740478
Sheet 1 of 1



1 - 1.10 to 5.53 m

DRAFT

DRAFT

AECOM Engineering Log

Borehole No. ABH8

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: JL
Checked by: PJW
Start Date: 06/05/26
End Date: 06/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 5.1 m

Hole Diameter: 200-96 mm
Inclination: 90°
Bearing: N/A

Easting: 338578.27
Northing: 6254377.81
Hor. Datum: MGA2020-56

RL: 14.345 m
Ver. Datum: AHD
Surface: Concrete

METHOD	SUPPORT	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
DT	U		Not Encountered	14.0	0.0			FILL - CONCRETE: dark grey, aggregates up to 20 mm, 480 mm thick, driveway surface.	D		CONCRETE LAYER
AS		B: 0.48-1.10 m			1.0		CL	Sandy CLAY: orange-brown and red-brown; sand, fine to medium grained.	M		EXTREMELY WEATHERED MATERIAL
				13.0							
					2.0						
				12.0							
					3.0						
				11.0							
					4.0						
				10.0							
					5.0						

Log continued on next page.

AECOM Engineering Log

DRAFT

Cored Borehole No. ABH8

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: JL
Checked by: PJW
Start Date: 06/05/26
End Date: 06/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 5.1 m

Hole Diameter: 200-96 mm
Inclination: 90°
Bearing: N/A

Easting: 338578.27
Northing: 6254377.81
Hor. Datum: MGA2020-56

RL: 14.345 m
Ver. Datum: AHD
Surface: Concrete

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x $Is_{(50)}$ A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
				14.0	0.0 – 1.0		*Log continued from previous page.*					
NMLC	1	Not Encountered	C: 1.43-1.53 m Is(50) D=0.58 MPa; C: 1.43-1.53 m Is(50) A=0.86 MPa	13.0	1.2 – 2.77		SANDSTONE: fine to medium grained, pale grey and orange-brown, iron stained.	HW; MW	78 (74) [39]			HAWKESBURY SANDSTONE 1.22-1.75 m: P, 0-5°, RF, PR, VN Fe, 20-100 mm spacing, x 8 2.12 m: DB 2.28-2.61 m: P, 0°, RF, PR, VN Qtz, 120-210 mm spacing, x 3 2.70 m: P, 0°, RF, IR, VN Qtz
			C: 2.40-2.50 m Is(50) D=0.74 MPa; C: 2.40-2.50 m Is(50) A=0.66 MPa	12.0	2.0							
					2.77 – 3.36		NO CORE: 2.77 to 3.36 m.					
	3.80		C: 3.70-3.80 m Is(50) D=0.64 MPa; C: 3.70-3.80 m Is(50) A=0.5 MPa	11.0	3.36 – 5.0		SANDSTONE: fine to medium grained, pale grey and red-brown, iron stained, bedded at 0-15°, 10-200 mm spacing, trace quartz inclusions.	HW; XW; HW; MW	100 (85) [82]			3.41-3.59 m: EW, 0°, RF, PR, Clay, 180 mm thick 3.70 m: P, 0°, RF, PR, VN Fe 3.93-3.96 m: MB, x 2 3.96 m: MB
	2		C: 4.57-4.65 m Is(50) D=0.93 MPa; C: 4.57-4.65 m Is(50) A=0.73 MPa; C: 4.62-4.97 m UCS=33	10.0	4.0 – 5.0							4.12 m: P, 0-25°, RF, PR, VN Fe, 50 mm spacing 4.15 m: DB 4.23 m: P, 0-25°, RF, PR, VN Fe, 50 mm spacing 4.38 m: P, 0-25°, RF, PR, VN Fe, 50 mm spacing 4.56 m: DB 4.58 m: P, 0-25°, RF, PR, VN Fe, 170 mm spacing 4.62 m: P, 0-25°, RF, PR, VN Fe, 60 mm spacing

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DRAFT

AECOM Engineering Log

Cored Borehole No. ABH8

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: JL
Checked by: PJW
Start Date: 06/05/26
End Date: 06/05/26
Location Meth.: dGPS0.1

Driller: BG Drilling
Drill Rig: Christie CE180
Total Depth: 5.1 m
Hole Diameter: 200-96 mm
Inclination: 90°
Bearing: N/A
Easting: 338578.27
Northing: 6254377.81
Hor. Datum: MGA2020-56
RL: 14.345 m
Ver. Datum: AHD
Surface: Concrete

METHOD	CORE RUN	GROUND WATER	SAMPLES & FIELD TESTS	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	MATERIAL DESCRIPTION (colour, grain size, texture and fabric, structure, bedding dip, soil moisture, consistency/density)	WEATHERING	TCR (SCR) [RQD] (%)	INFERRED STRENGTH 20 x Is(50) A: ● D: o UCS: □ (SOIL 0-6, VL 2, L 6, M 20, H 60, VH 200, EH)	DEFECT SPACING (mm) (20, 60, 200, 600, 2000)	ADDITIONAL OBSERVATIONS (Defect Descriptions)
NMLC		Not Encounter	MPa C: 5.00-5.10 m Is(50) D=0.33 MPa C: 5.00-5.10 m Is(50) A=0.25 MPa				SANDSTONE: fine to medium grained, pale grey and red-brown, iron stained, bedded at 0-15°, 10-200 mm spacing, trace quartz inclusions. Terminated at 5.10 m. Target depth.	MW		●o		5.07 m: DB

RL (m AHD) scale: 9.0, 8.0, 7.0, 6.0, 5.0

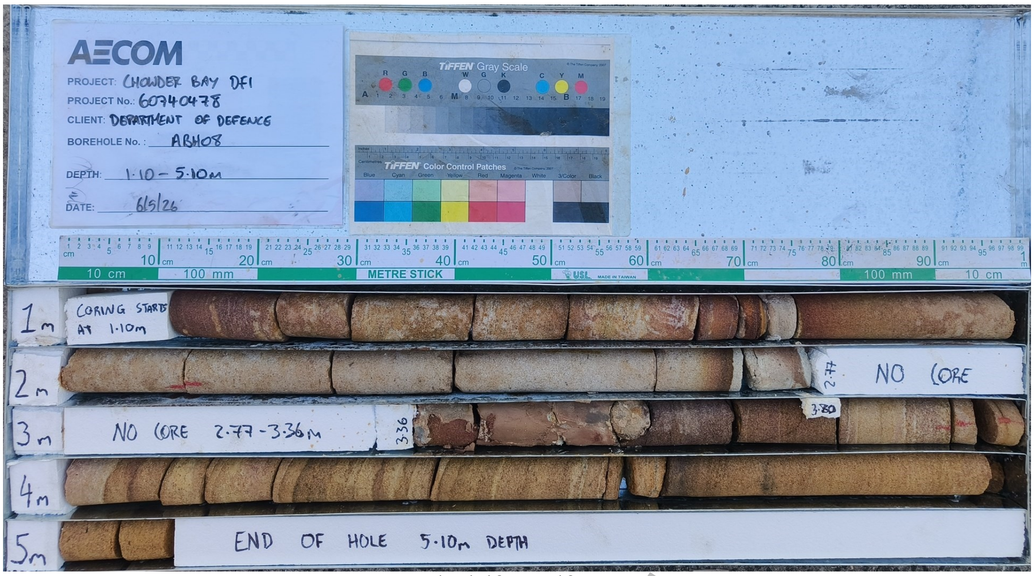
DEPTH (m) scale: 6.0, 7.0, 8.0, 9.0, 10.0

CORE PHOTO REPORT

ID NO. : ABH8

PROJECT : Chowder Bay DFI
LOCATION : Chowder Bay

PROJECT NO. : 60740478
Sheet 1 of 1



1 - 1.10 to 5.10 m

DRAFT

AECOM Engineering Log

Borehole No. HA1

Sheet 1 of 1

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 07/05/26
End Date: 07/05/26
Location Meth.: dGPS0.1

Driller: AECOM
Drill Rig: Hand Auger
Total Depth: 0.2 m

Hole Diameter: 75 mm
Inclination: 90°
Bearing: N/A

Easting: 338520.97
Northing: 6254463.88
Hor. Datum: MGA2020-56

RL: 22.288 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	1 21db	D: 0.00-0.20 m	Not Encounter		0.0			TOPSOIL - Clayey SAND: fine to medium grained, dark brown; with gravel, fine to coarse, sub-rounded, sandstone; trace cobbles, size up to 70mm, sub-rounded, sandstone.	D		TOPSOIL
									Terminated at 0.20 m. Hand auger refusal.			
					22.0							
						1.0						
					21.0							
						2.0						
					20.0							
						3.0						
					19.0							
						4.0						
					18.0							
						5.0						

DRAFT

AECOM Engineering Log

Borehole No. HA2

Sheet 1 of 1

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 08/05/26
End Date: 08/05/26
Location Meth.: dGPS0.1

Driller: AECOM
Drill Rig: Hand Auger
Total Depth: 1 m

Hole Diameter: 75 mm
Inclination: 90°
Bearing: N/A

Easting: 338596.28
Northing: 6254430.27
Hor. Datum: MGA2020-56

RL: 30.448 m
Ver. Datum: AHD
Surface: Topsoil

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	1, 4, 4	D: 0.00-0.30 m ES: 0.00-0.30 m	Not Encountered		0.0			TOPSOIL - Clayey SAND: fine to medium grained, dark brown; trace rootlets.	D		TOPSOIL
		3	D: 0.30-0.40 m ES: 0.30-0.40 m					SC	Clayey SAND: fine to medium grained, yellow-brown; trace gravel, fine to medium, sub-rounded, sandstone.	D	MD	RESIDUAL SOIL
		3, 4, 4, 9	D: 0.40-0.80 m ES: 0.40-0.80 m		30.0			CH	Sandy Silty CLAY: high plasticity, brown-grey mottled orange; sand, fine grained; trace gravel, fine to medium, sub-rounded, sandstone.	w ≈ PL	St	
		22	D: 0.80-1.00 m ES: 0.80-1.00 m						From 0.8m, pale grey mottled orange	w < PL	H	
						1.0			Terminated at 1.00 m. Hand auger refusal.			
					29.0							
						2.0						
					28.0							
						3.0						
					27.0							
						4.0						
					26.0							
						5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DRAFT

AECOM Engineering Log

Borehole No. HA3

Sheet 1 of 1

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 08/05/26
End Date: 08/05/26
Location Meth.: dGPS0.1

Driller: AECOM
Drill Rig: Hand Auger
Total Depth: 0.8 m

Hole Diameter: 75 mm
Inclination: 90°
Bearing: N/A

Easting: 338602.83
Northing: 6254396.12
Hor. Datum: MGA2020-56

RL: 19.001 m
Ver. Datum: AHD
Surface: Fill

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	2, 5, 7, 7	D: 0.00-0.40 m ES: 0.00-0.40 m	Not Encountered	19.0	0.0			FILL - Clayey SAND: fine to medium grained, dark brown; trace tree roots.	D		FILL
		6, 7, 7, 8	D: 0.40-0.80 m ES: 0.40-0.80 m					CI	Sandy Silty CLAY: medium plasticity, yellow-brown; sand, fine to medium grained; trace gravel, fine to medium, sub-rounded, sandstone.	w < PL	VSt	RESIDUAL SOIL
		12, 18, 22			18.0	1.0			Terminated at 0.80 m. Hand auger refusal.			
					17.0	2.0						
					16.0	3.0						
					15.0	4.0						
						5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DRAFT

AECOM **Engineering Log**

Borehole No. HA4

Sheet 1 of 1

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 08/05/26
End Date: 08/05/26
Location Meth.: dGPS0.1

Driller: AECOM
Drill Rig: Hand Auger
Total Depth: 0.3 m
Hole Diameter: 75 mm
Inclination: 90°
Bearing: N/A
Easting: 338566.80
Northing: 6254384.66
Hor. Datum: MGA2020-56
RL: 14.325 m
Ver. Datum: AHD
Surface: Fill

METHOD	SUPPORT	DCP (BLOWS PER 100 mm)	SAMPLES & FIELD TESTS	GROUND WATER	RL (m AHD)	DEPTH (m)	GRAPHIC LOG	CLASSIFICATION SYMBOL	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	MOISTURE CONDITION	CONSISTENCY / RELATIVE DENSITY	ADDITIONAL OBSERVATIONS (Geological Origin)
HA	U	1 2 15db	D: 0.00-0.30 m ES: 0.00-0.30 m	Not Encounter		0.0			FILL - Clayey SAND: fine to medium grained, dark brown.	D		FILL
					14.0				Terminated at 0.30 m. Hand auger refusal.			
						1.0						
					13.0							
						2.0						
					12.0							
						3.0						
					11.0							
						4.0						
					10.0							
						5.0						

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DRAFT

Engineering Log

Piezometer No. ABH1/MW008

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay Road, Chowder Bay, NSW, Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 29/04/26
End Date: 30/04/26

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 13.05 m

Hole Diameter: 125 mm
Inclination: 90°
Bearing: N/A

Easting: 338457.54
Northing: 6254468.30
Hor. Datum: MGA2020-56

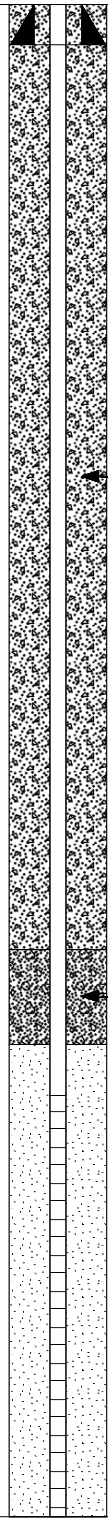
RL: 26.1 m
Ver Datum: AHD
Surface: Topsoil

METHOD	GROUND WATER	SAMPLES & FIELD TESTS	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)
HA		B: 0.00-0.30 m	TOPSOIL - Clayey SAND: fine to medium grained, dark brown; trace gravel, fine to medium, sub-rounded, sandstone .
			FILL - Clayey SAND: fine to medium grained, brown mottled yellow; trace gravel, fine to coarse, sub-rounded, sandstone ; trace cobbles, size up to 64mm, sub-rounded, sandstone .
			FILL - MASONRY: pale grey, with visible voids; bricks with mortar joints.
			NO CORE: from 0.50 to 1.10m.
		C: 1.28-1.39 m Is(50) D=3 MPa C: 1.28-1.39 m Is(50) A=1.9 MPa	FILL - MASONRY: brown to reddish brown; bricks with mortrr joints.
			NO CORE: from 1.52 to 2.20m.
		C: 2.20-2.31 m Is(50) D=0.46 MPa C: 2.20-2.31 m Is(50) A=0.99 MPa C: 3.13-3.23 m Is(50) D=1.8 MPa C: 3.13-3.23 m Is(50) A=0.34 MPa	FILL - MASONRY: grey to reddish brown; bricks with mortar joints.
			NO CORE: from 3.77 to 4.20m.
		C: 4.63-4.73 m Is(50) D=0.6 MPa C: 4.63-4.73 m Is(50) A=0.53 MPa	FILL - MASONRY: reddish brown; bricks with mortar joints.
			NO CORE: from 4.83 to 5.50m.
			ROOT: black.
		C: 5.79-5.87 m Is(50) D=0.16 MPa C: 5.79-5.87 m Is(50) A=0.28 MPa	SANDSTONE: fine to coarse grained, brown, indistinctly bedded.
	08/05/2026 13:07:43		SANDSTONE: medium to coarse grained, red-brown mottled grey, indistinctly bedded.
		C: 6.75-6.84 m Is(50) D=0.66 MPa C: 6.75-6.84 m Is(50) A=0.83 MPa	SANDSTONE: fine to medium grained, pale grey, cross-bedded at 0-15°, with dark grey laminations .
		C: 7.79-7.88 m Is(50)	SANDSTONE: fine with medium grained, brown-grey, with white quartz inclusions, sub-rounded, size up to 5mm.

RL (m AHD): 26.0, 25.0, 24.0, 23.0, 22.0, 21.0, 20.0, 19.0

DEPTH (m): 0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0

WELL DIAGRAM



AECOM **Engineering Log**

DRAFT

Piezometer No. ABH1/MW008

Client: Department of Defence		**Project No:** 60740478	
Project: Chowder Bay DFI		**Logged by:** AL	**Checked by:** PJW
Location: Chowder Bay Road, Chowder Bay, NSW, Chowder Bay		**Start Date:** 29/04/26	**End Date:** 30/04/26
Driller: Hard Access Drilling	**Hole Diameter:** 125 mm	**Easting:** 338457.54	**RL:** 26.1 m
Drill Rig: Man Portable Rig	**Inclination:** 90°	**Northing:** 6254468.30	**Ver Datum:** AHD
Total Depth: 13.05 m	**Bearing:** N/A	**Hor. Datum:** MGA2020-56	**Surface:** Topsoil

METHOD	GROUND WATER	SAMPLES & FIELD TESTS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD)	DEPTH (m)	WELL DIAGRAM
		D=0.77 MPa C: 7.79-7.88 m Is(50) A=1.1 MPa C: 8.18-8.27 m Is(50) D=0.14 MPa C: 8.18-8.27 m Is(50) A=0.1 MPa		SANDSTONE: fine with medium grained, brown-grey, with white quartz inclusions, sub-rounded, size up to 5mm. From 8.26m, orange brown, iron stained	18.0		
		UCS = 22 MPa C: 9.00-9.20 m C: 9.20-9.29 m Is(50) D=0.94 MPa C: 9.20-9.29 m Is(50) A=1.1 MPa C: 10.57-10.66 m Is(50) D=1.3 MPa C: 10.57-10.66 m Is(50) A=1.2 MPa C: 11.57-11.65 m Is(50) D=0.47 MPa		SANDSTONE: fine to medium grained, pale grey mottled brown, cross-bedded at 0-20°, with dark grey laminations .	17.0 16.0 15.0	9.0 10.0 11.0	Sand
		C: 11.57-11.65 m Is(50) A=0.85 MPa		SANDSTONE: fine to medium grained, pale grey, indistinctly bedded, with black flecks.	14.0	12.0	
		C: 12.41-12.50 m Is(50) D=1.3 MPa C: 12.41-12.50 m Is(50) A=0.67 MPa		SANDSTONE: fine to medium grained, pale grey mottled black flecks, cross-bedded at 0-10° .		13.0	
				Terminated at 13.05 m. Target depth.	13.0 12.0 11.0	14.0 15.0 16.0	

Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DRAFT

AECOM Engineering Log

Piezometer No. ABH2/MW009

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay Road, Chowder Bay, NSW, Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 06/05/26
End Date: 06/05/26

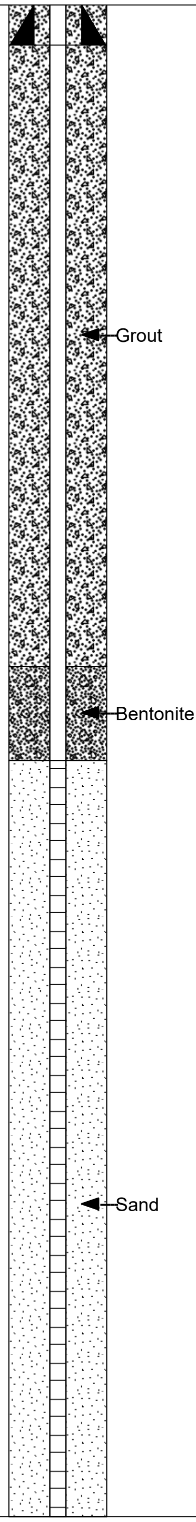
Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 8.7 m
Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A
Easting: 338511.26
Northing: 6254456.34
Hor. Datum: MGA2020-56
RL: 19.878 m
Ver Datum: AHD
Surface: Concrete

METHOD	GROUND WATER	SAMPLES & FIELD TESTS	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)
DT		CONCC: 0.00-0.12 m ES: 0.00-0.15 m	FILL - CONCRETE: pale grey, aggregates size up to 25mm, with visible voids.
HA		D: 0.12-0.33 m	FILL - Gravelly SAND: fine to coarse grained, dark brown; gravel, fine to coarse, sub-rounded, sandstone.
AD/T		D: 0.33-1.00 m	FILL - Clayey SAND: fine to coarse grained, brown; trace gravel, fine to medium, sub-angular to sub-rounded, sandstone .
AD/T		D: 1.00-2.00 m ES: 1.00-2.00 m SPT: 2.00-2.45 m 1,1,1 N=2 D: 2.00-2.45 m D: 2.00-3.00 m SPT: 3.50-3.95 m 3,3,2 N=5 D: 3.50-3.95 m SPT: 4.00-4.45 m 8,3,7 N=10 ES: 4.00-4.45 m	FILL - Silty Gravelly SAND: fine to coarse grained, yellow-brown mottled grey; gravel, fine to medium, sub-rounded, sandstone .
	08/05/2026 14:09:21	C: 4.72-4.77 m Is(50) A=0.26 MPa C: 4.72-4.77 m	SANDSTONE: medium to coarse grained, yellow-brown mottled red, indistinctly bedded, trace dark brown laminations and iron staining .
			NO CORE: from 4.77 to 5.68m.
		C: 5.76-5.89 m Is(50) D=0.39 MPa C: 5.76-5.89 m Is(50) A=0.5 MPa	SANDSTONE: fine to coarse grained, brown mottled grey, cross-bedded at 0-30°.
			SANDSTONE: fine to medium grained, pale grey mottled brown, indistinctly bedded, with dark grey flecks and iron stained bands.
		C: 6.72-6.82 m Is(50) D=0.69 MPa C: 6.72-6.82 m Is(50) A=0.69 MPa	SANDSTONE: fine to coarse grained, brown-grey, cross-bedded at 0-15°, with brown laminations and quartz inclusions, trace iron stained bands.
		C: 7.54-7.65 m Is(50) D=0.4 MPa C: 7.54-7.65 m Is(50)	SANDSTONE: fine to medium grained, dark brown mottled grey, cross-bedded at 0-20°, with quartz inclusions and dark brown laminations.

RL (m AHD): 19.0, 18.0, 17.0, 16.0, 15.0, 14.0, 13.0, 12.0

DEPTH (m): 0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0

WELL DIAGRAM



Engineering log should be read in conjunction with AECOM soil and rock logging description sheets.

DRAFT

Engineering Log

Piezometer No. ABH2/MW009

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay Road, Chowder Bay, NSW, Chowder Bay

Project No: 60740478
Logged by: AL
Checked by: PJW
Start Date: 06/05/26
End Date: 06/05/26

Driller: Hard Access Drilling
Drill Rig: Man Portable Rig
Total Depth: 8.7 m

Hole Diameter: 96 mm
Inclination: 90°
Bearing: N/A

Easting: 338511.26
Northing: 6254456.34
Hor. Datum: MGA2020-56

RL: 19.878 m
Ver Datum: AHD
Surface: Concrete

METHOD	GROUND WATER	SAMPLES & FIELD TESTS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD)	DEPTH (m)	WELL DIAGRAM
		A=0.59 MPa C: 8.28-8.37 m Is(50) D=0.59 MPa		SANDSTONE: fine to coarse grained, orange-brown mottled grey, cross-bedded at 0-15°, with quartz inclusions and dark grey flecks.			
		C: 8.28-8.37 m Is(50) A=0.77 MPa		SANDSTONE: fine to medium grained, dark grey, indistinctly bedded, with quartz inclusions and iron stained bands.			
				Terminated at 8.70 m. Target depth.	11.0		
						9.0	
					10.0		
						10.0	
					9.0		
						11.0	
					8.0		
						12.0	
					7.0		
						13.0	
					6.0		
						14.0	
					5.0		
						15.0	
					4.0		
						16.0	

DRAFT

AECOM Engineering Log

Piezometer No. ABH6/MW010

Sheet 1 of 1

Client: Department of Defence
Project: Chowder Bay DFI
Location: Chowder Bay Road, Chowder Bay, NSW, Chowder Bay

Project No: 60740478
Logged by: JL
Checked by: PJW
Start Date: 08/05/26
End Date: 08/05/26

Driller:
Drill Rig: Christie CE180
Total Depth: 5.88 m
Hole Diameter: 125 mm
Inclination: 90°
Bearing: N/A
Easting: 338571.37
Northing: 6254415.13
Hor. Datum: MGA2020-56
RL: 27.311 m
Ver Datum: AHD
Surface: Concrete

METHOD	GROUND WATER	SAMPLES & FIELD TESTS	GRAPHIC LOG	MATERIAL DESCRIPTION (plasticity/particle characteristics, colour, secondary and other minor components, structure)	RL (m AHD)	DEPTH (m)	WELL DIAGRAM
DT		D: 0.06-0.50 m		FILL - CONCRETE: dark grey, aggregates up to 10 mm diameter, 60 mm thick; road surface.		0.0	Concrete
AD/T		D: 0.50-1.50 m		FILL - Gravelly SAND: fine to coarse grained, dark grey; gravel, fine to medium, sub-angular, sandstone; trace clay.	27.0		Bentonite
				CL Sandy CLAY: low plasticity, pale grey and brown; sand, fine to coarse grained.	26.0	1.0	
		SPT: 1.50-1.54 m 25/40 mm HB N=R C: 1.64-1.73 m Is(50) D=0.094 MPa C: 1.64-1.73 m Is(50) A=0.11 MPa		SANDSTONE: fine to medium grained, pale grey, indistinctly bedded, with carbonaceous laminations and flecks, trace iron staining , trace quartz inclusions.		2.0	
		C: 2.35-2.44 m Is(50) D=0.074 MPa C: 2.35-2.44 m Is(50) A=0.18 MPa		from 2.74 to 3.00 m: bedded at 0-5°, variable spacing	25.0	3.0	
		C: 3.24-3.32 m Is(50) D=0.21 MPa C: 3.24-3.32 m Is(50) A=0.19 MPa			24.0	4.0	Sand
		C: 4.45-4.55 m Is(50) D=0.13 MPa C: 4.45-4.55 m Is(50) A=0.11 MPa			23.0	5.0	
		C: 5.50-5.58 m Is(50) D=0.092 MPa C: 5.50-5.58 m Is(50) A=0.083 MPa			22.0		
				Terminated at 5.88 m. Target depth.	21.0	6.0	
					20.0	7.0	
						8.0	